

Efficient Split Learning

Optimizing Communication and Computation in Federated Learning for Resource-Constrained Devices

MTech Thesis Presentation



Manisha Chawla (12210370)

MTech CSE(2022-24)

Under the supervision of

Dr. Gagan Raj Gupta

Indian Institute Of Technology, Bhilai

Department of Computer Science & Engineering

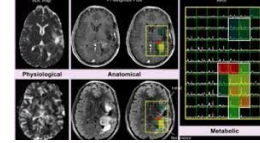
May 01, 2024

Presentation Outline

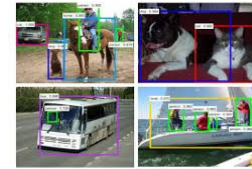
- Research Background
- Literature Review
- Research Gap
- Motivation
- ESL System Architecture
- ESL Algorithm
- Experimental Goals
- Experimental Setup
- Results and Use Case in Medical Imaging
- Conclusion
- Future Work

Research Background: Communicate to Learn

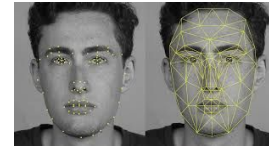
- Deep Learning models thrive on large datasets.
- Lack of samples at single location.
- Data privacy.



I. Medical Imaging



II. Object Detection



III. Image Recognition

This motivates to develop communication efficient, privacy preserving distributed learning on resource constrained devices.

[I] https://bids.berkeley.edu/sites/default/files/styles/400x225/public/projects/vareth_-_nmr_cover_new_project_page_banner.png?itok=FGmASmU2
[II] <https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcSNcllVUIYAa5IQN1wiSbJbdcowkH84NFacPA>
[III] https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQwjns0JqaFaxnXG_zbRtFFVzOfco1ulxvpgQ

Research Background: Federated Learning

Benefits:

- Data Privacy
- Scalable Architecture

Limitation:

- Requires heavy resource consumption
- Disparities in data distribution
- Communication Overhead

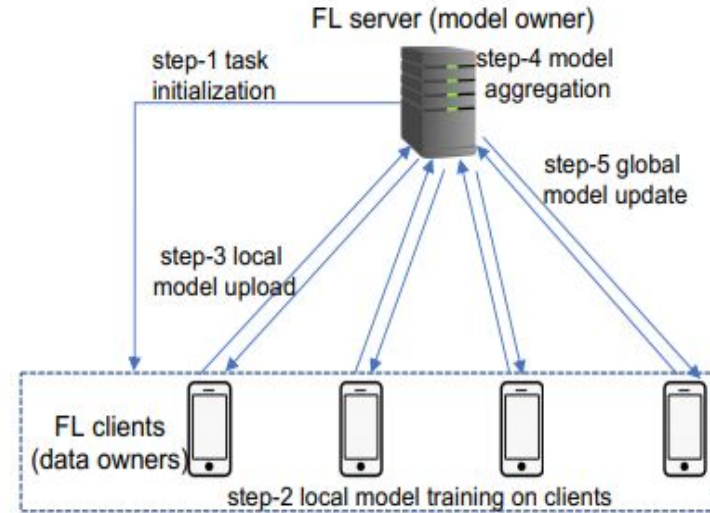


Figure: The framework architecture of Federated Learning.

This motivated the development of Split Learning Approaches

Research Background: Split Learning (SL)

Benefits:

- Data Privacy
- Scalable Architecture and Edge Computing Benefits
- Reduces Computation overhead

Limitation:

- Disparities in data distribution
- Introduces additional communication overheads.

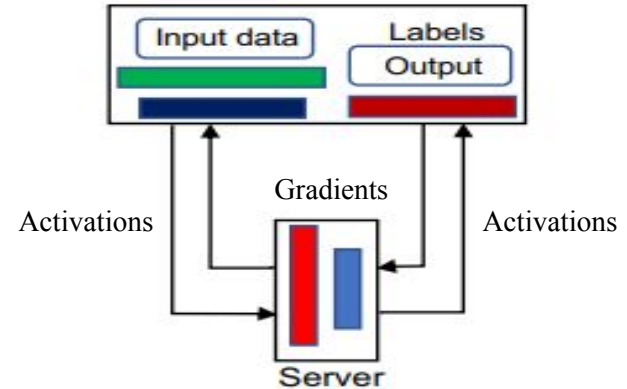


Figure: Split Learning without Label sharing

Therefore, communication efficiency is an important aspect of SL performance.

Literature Review:

Traditional approach to reduce communication overhead:

- Accelerate Convergence Strategies:
 - Momentum-based Optimization[1]
 - Asynchronous Training: [8][9]
- Communication Compression Techniques:
 - Sparsification, Quantization, and Pruning[10][11][12],
 - Model Compression[13][14][15]
 - Autoencoder-based Compression[16][17][18]

Traditional Approach tackle data heterogeneity:

- Enhance FL Algorithm[1][2][3]
- Adaptive Learning Rates[4]
- Personalised Learning[5][6][7]

Literature Review: Accelerate Convergence Strategies

S. No	Authors with Paper Title / Year of Publication	Technical Analysis / Brief of the Paper	Conference/Journal/ Transaction
[1]	Brendan McMahan et. al. FedAvg: Communication Efficient Learning of Deep Networks from Decentralized Data /2017	- It performs iterative round-based training. - Each round consists of local mini-batch updates on each client followed by parameter averaging on a central server. - This strategy is known to be sensitive to heterogeneity when the number of local updates grows.	In Artificial intelligence and statistics, pages 1273–1282.
[8]	Yusuke Kanamori et. al. An asynchronous federated learning focusing on updated models for decentralized systems with a practical framework / 2023	- To aggregate only the updated models after the previous round to improve the quality of the device's model by avoiding the reaggregation of older models	2023 IEEE 47th Annual Computers, Software, and Applications Conference (COMPSAC)
[9]	Xing Chen et. al. Communication and Computation Reduction for Split Learning using Asynchronous Training / 2021	- Loss-based asynchronous training scheme that updates the client-side model less frequently and only sends/receives activations/gradients in selected epochs.	IEEE Workshop on Signal Processing Systems (SiPS)

Literature Review: Communication Compression Techniques



S. No	Authors with Paper Title / Year of Publication	Technical Analysis / Brief of the Paper	Conference/Journal/ Transaction
[10]	Pengchao Han et. al. Adaptive Gradient Sparsification for Efficient Federated Learning/ 2020	- Gradient Sparsification (GS) can be applied, where instead of the full gradient, only a small subset of important elements of the gradient is communicated.	IEEE 40th International Conference on Distributed Computing Systems (ICDCS)
[13]	Felix Sattler et. al. Robust and Communication-Efficient Federated Learning from Non-IID Data / 2019	- Enable downstream compression as well as ternarization and optimal Golomb encoding of the weight updates.	IEEE Transactions on Neural Networks and Learning Systems
[16]	Mahdi Beitollahi et. al. FLAC: Federated Learning with Autoencoder Compression and Convergence Guarantee / 2022	- Compress user devices' models for uplink transmission	IEEE Global Communications Conference
[11]	Fei Zheng et. al. Reducing Communication for Split Learning by Randomized Top-k Sparsification /2023	- Selecting top-k elements with a large probability while also having a small probability to select non-top-k elements.	International Joint Conference on Artificial Intelligence(IJCAT)

Literature Review: Addressing Heterogeneity in FL

S. No	Authors with Paper Title / Year of Publication	Technical Analysis / Brief of the Paper	Conference/Journal/ Transaction
[1]	Brendan McMahan et. al. FedAvg: Communication Efficient Learning of Deep Networks from Decentralized Data /2017	- It performs iterative round-based training. - Each round consists of local mini-batch updates on each client followed by parameter averaging on a central server. - This strategy is known to be sensitive to heterogeneity when the number of local updates grows.	In Artificial intelligence and statistics, pages 1273–1282.
[2]	Tian Li et. al. FedProx: Federated optimization in heterogeneous networks /2020	- To mitigate statistical heterogeneity, FedProx builds on FedAvg by introducing a regularization term to each local training loss. - Thereby controlling the deviation of the local models from the last global model.	Proceedings of Machine Learning and Systems, 2:429–450.
[3]	Sai Praneeth Karimireddy et. al. Scaffold: Stochastic controlled averaging for federated learning / 2020	- Scaffold mitigates client drifts using control-variates and by adding a server-side learning rate.	In International Conference on Machine Learning, pages 5132–514.

Research Gap:

Challenges with Traditional Approach:

❑ Trade-offs:

- ❑ Traditional strategies prioritize compression or fast convergence techniques to reduced communication. It affects model accuracy.
- ❑ For example “Communication for Split Learning by Randomized Top-k Sparsification, 2023[11]” resulted in an 88% reduction in communication but caused a 3% decrease in accuracy.

❑ Unaddressed Concern:

- ❑ Computation overhead remains a critical issue in existing FL/SL approaches.
- ❑ The experimental study lacked diversity in complex tasks and realistic datasets.

Problem Statement

In our study, we seek to achieve optimal performance, especially in non-IID settings, with a focus on rapid convergence, minimized computation and communication overhead using Split Learning Setting.

Efficient Split Learning: Main Idea



1. Deep Learning model as a function

$$f(x) = y$$

2. Transfer Learning

$$f(x) = f_B f_F(x)$$

3. U Shape Split

$$f(x) = f_{CB} f_{SB} f_{SF} f_{CF}(x)$$

4. Key Value Store

$$\text{key}(x) = \text{unique_key}(\text{client_id}, \text{data_id})$$

$$\text{value}_{\text{server}}(x) = f_{SF} f_{CF}(x)$$

$$f(x) = f_{CB} f_{SB}(\text{value}_{\text{server}}(x))$$

5. Model Customization

$$f(x) = f_{\text{Custome_Block}_B} f_B(\text{value}(\text{key}(x)))$$

6. Model aggregation

$$f_{\text{Gen}}(x) = \sum_{\forall \text{ clients}} f_{\text{Custome_Block}_B} f_B(\text{value}_{\text{server}}(\text{key}(x)))$$

7. Personalised Training

$$\text{value}_{\text{client}_i}(x) = f_B(\text{value}_{\text{server}}(x_{\text{client}_i}))$$

$$f_{\text{Personal}_i}(x) = f_{\text{Personal_Custome_Block}}(\text{value}_{\text{client}_i}(x_{\text{client}_i}))$$

ESL Framework:

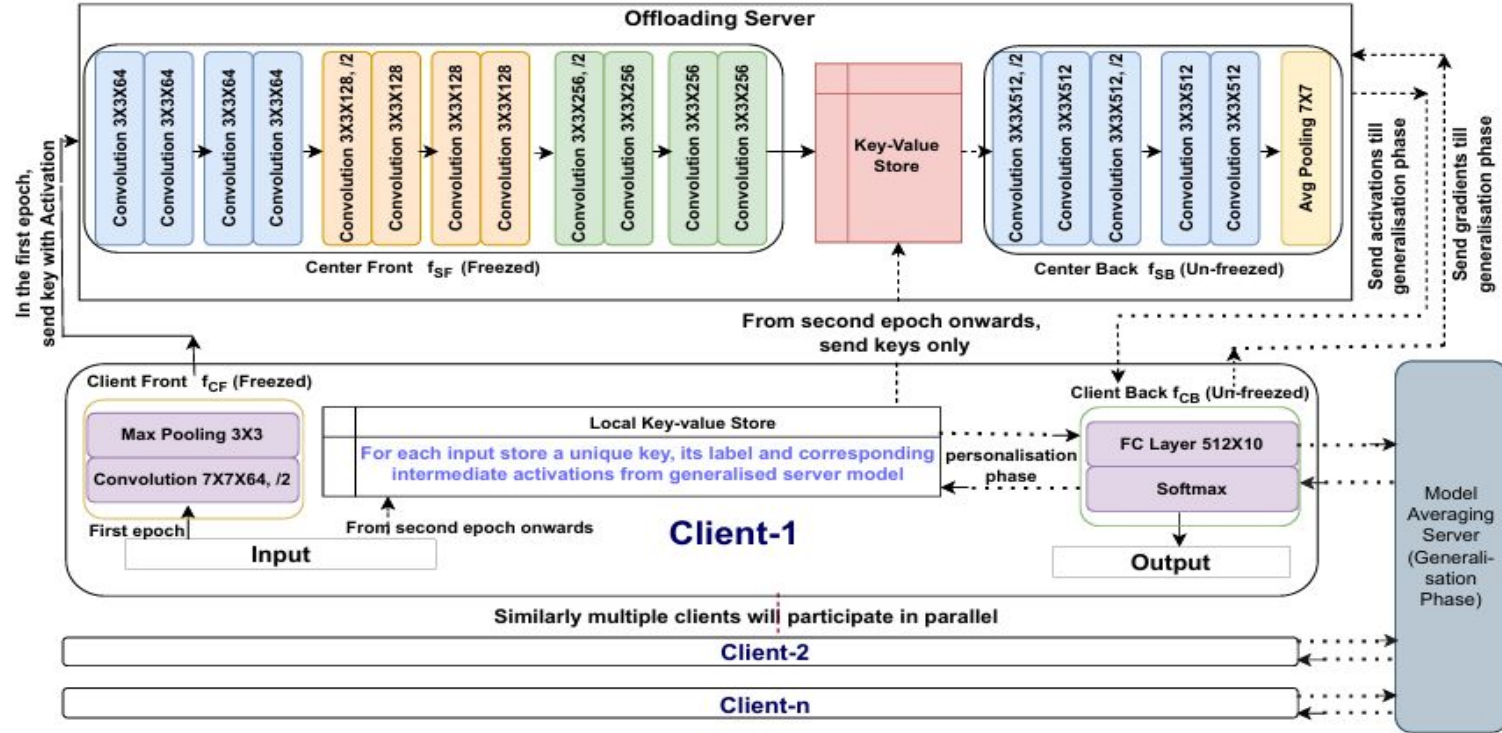


Figure: ESL System Architecture using ResNet18

As an example for splitting the model into the client front, center front, center back, and client back models for training with key-value store.

ESL Generalisation Personalisation Model:

Generalization Phase: Outputs common model working on all data distributions

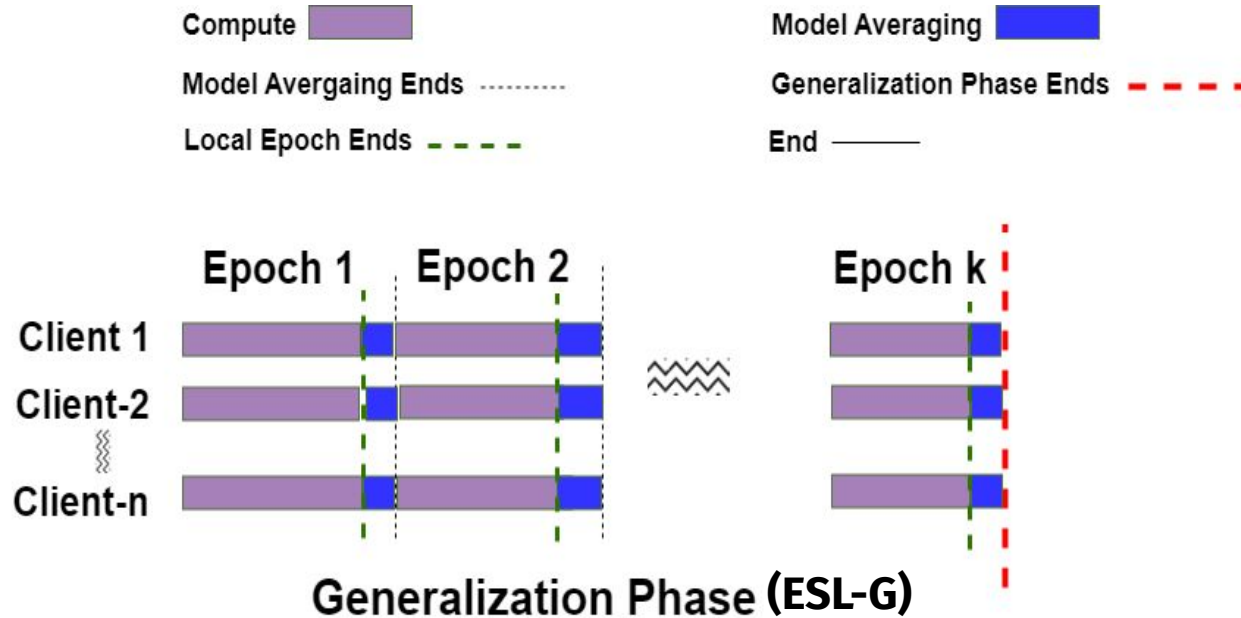


Figure: Training with multiple clients in ESL

ESL Generalisation Personalisation Model:

Generalization Phase: Outputs common model working on all data distributions.

Personalization Phase: Outputs models working best on client's own local data distribution

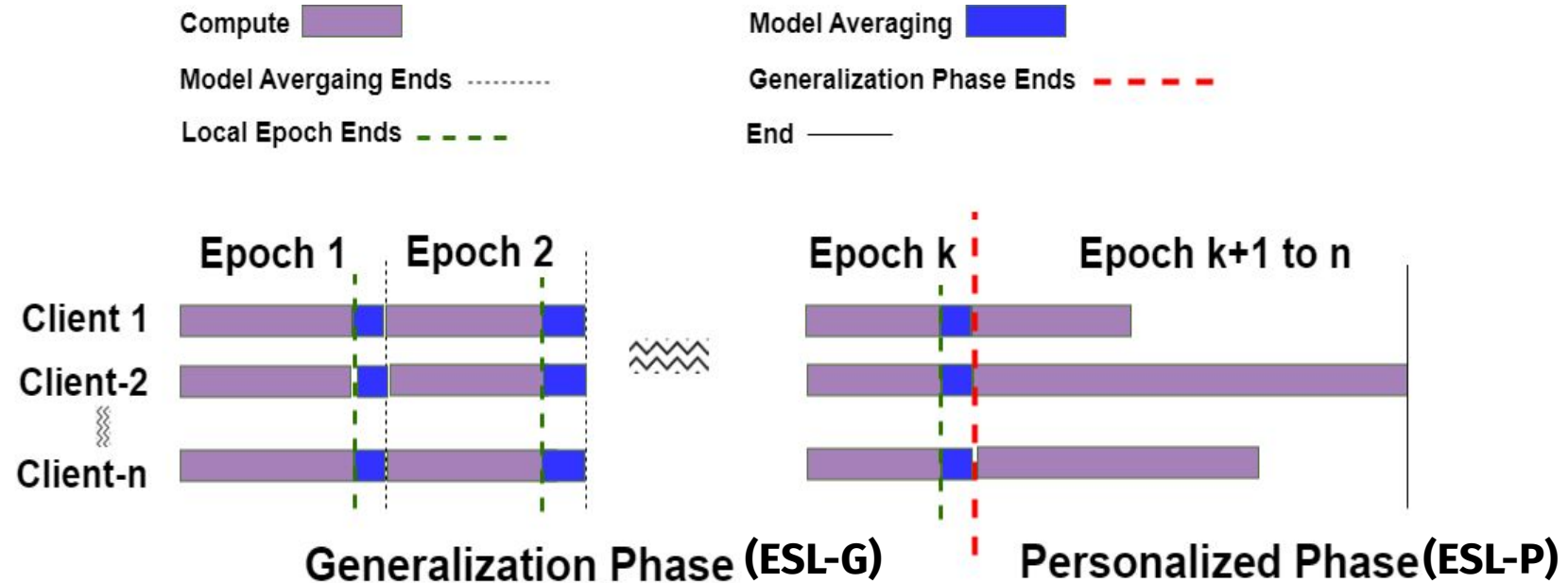


Figure: Training with multiple clients in ESL

Summary: Efficient Split Learning(ESL)

Challenges with Decentralised Training	Key-Value Store	Base Model Customization	Generalization-Personalization Model	Transfer Learning	U-shaped Configuration	Client Model Averaging Server
Statistical Heterogeneity		✓	✓			
Communication Overheads	✓		✓	✓		
Constrained Resources	✓	✓	✓	✓		
Convergence			✓	✓		
Privacy & Security					✓	✓

The main challenges to SL in IoT and representative **contribution** and **technical strategies** for addressing them.

Experiment Goal:

We have designed the experiments to answer the following questions:

1. How does varying the number of layers in the client's backside model affect SL performance and the computational workload for resource-constrained clients.
2. How does customizing the model at the client back side affect SL performance and workload distribution between clients and servers?
3. What is the overall performance of ESL compared to other existing algorithms for image classification and 3D image segmentation?
4. Is the key-value store approach effective in ESL, and how much does it mitigate overhead in terms of communication and computation compared to other methods?
5. Will the ESL framework demonstrate effective performance when applied to realistic medical datasets in a federated setting?
 - **Skin Lesion Diagnosis using ESL Framework**
 - **Diabetic Retinopathy Detection using ESL Framework**
 - **Brain 3D Image Segmentation using ESL framework**
 - **Kidney and Tumor Segmentation using ESL framework**

Experimental Setup:

Dataset Used:

Dataset	Number of Classes	Metric Used	Task	Base Model	Pretrained On Dataset
CIFAR-10[19] FMNIST[20]	10 10	Accuracy	Image Classification	ResNet-18[26]	ImageNet[30]
DR[21][22] ISIC-2019[23]	3 8	Balanced Accuracy[27]	Medical-Image Classification	ResNet-18	ImageNet
KITS-19[24] IXI-Tiny[25]	3 2	DICE-Score[28]	3D-Image Segmentation	nnUNet[31] 3D UNet[32]	MSD Pancreas[33] MSD Spleen[34]

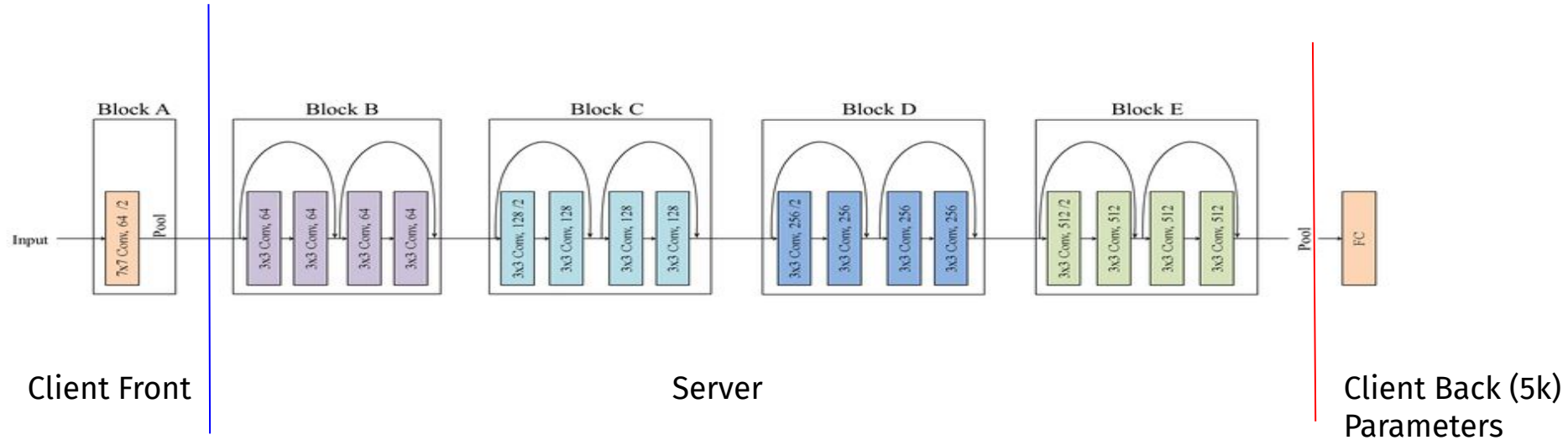
Framework Used: We have created our framework using PyTorch.

Experiments Conducted on: Workstation equipped with an Intel(R) Xeon(R) Gold 5218 CPU operating at 2.30GHz, coupled with the NVIDIA RTX A6000 GPU.

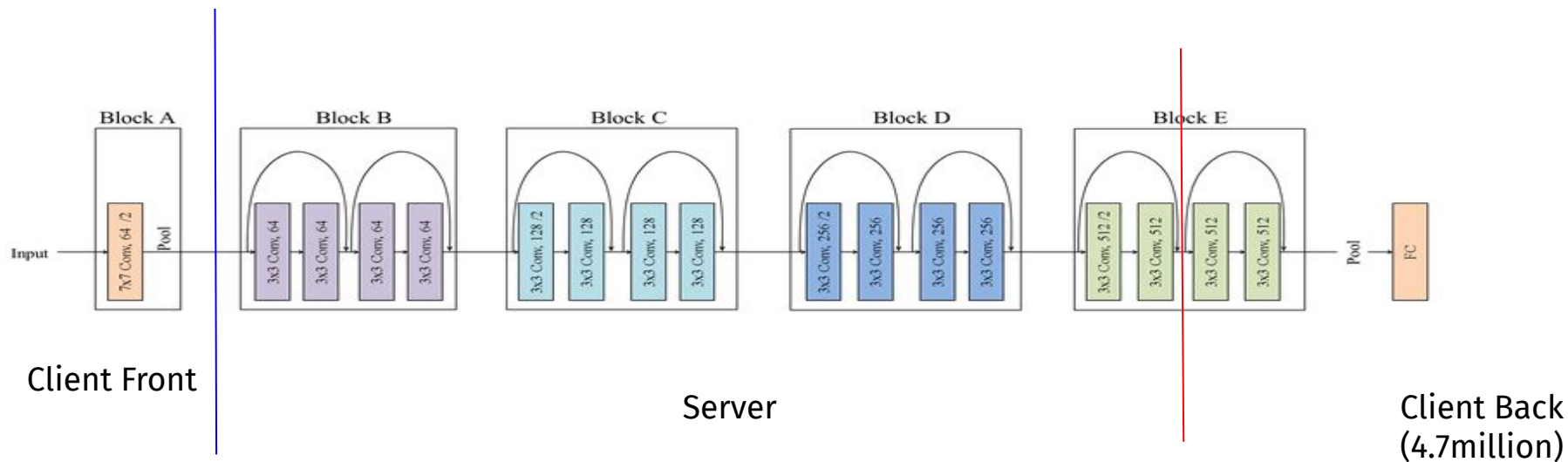
Reproducibility: Each experiment is repeated 5 times and average value is reported.

1. Effect of number of layers in client-back side.

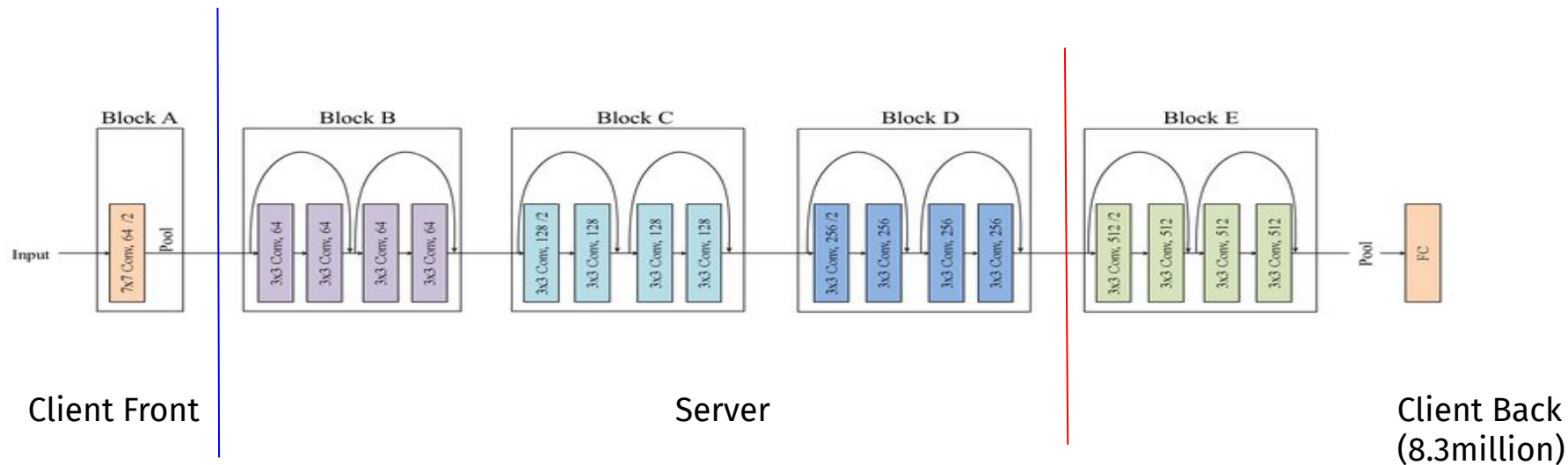
Setup: Splits-1



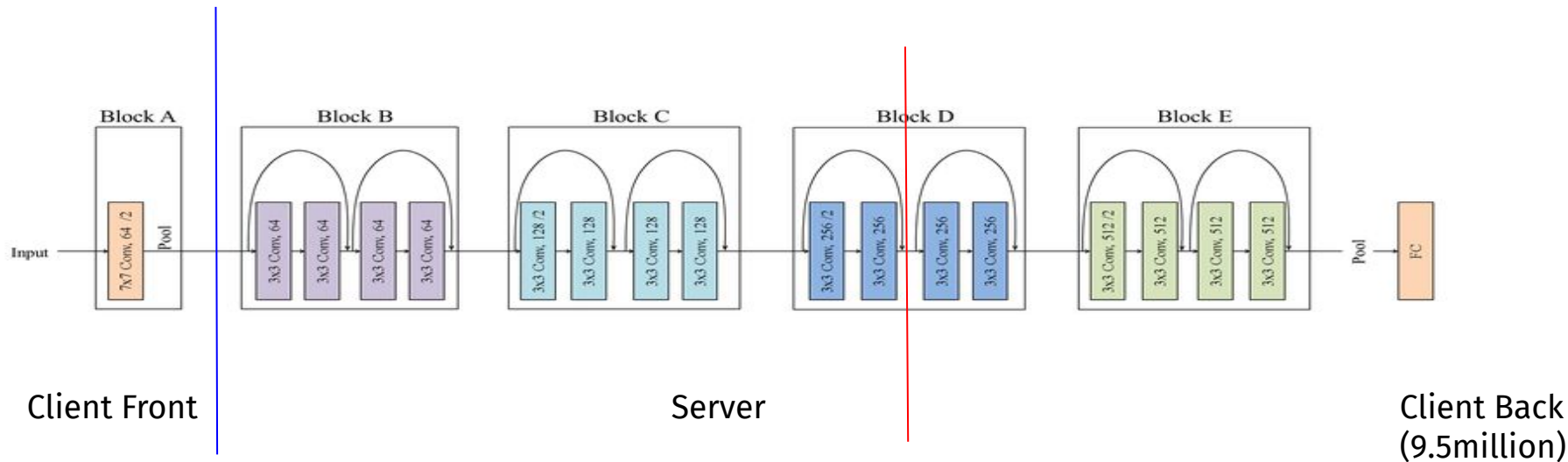
Setup: Splits-2



Setup: Splits-3



Setup: Splits-4



IID Setting: Different Model Split Results

small →

Large →

Model Split	Number of Parameters at Client Back (in million)	E-1		E-2	
		Test Accuracy	Epoch	Test Accuracy	Epoch
RN-S1	0.005	88.19	33	74.92	16
RN-S2	4.7	88.31	34	74.94	17
RN-S3	8.3	88.28	36	75.15	14
RN-S4	9.5	88.23	33	75.35	14

Performance metrics (Test Accuracy and Epoch) for different splits in an IID setting: E-1 (500 datapoints) and E-2 (50 datapoints) on CIFAR-10 dataset with 10 clients.

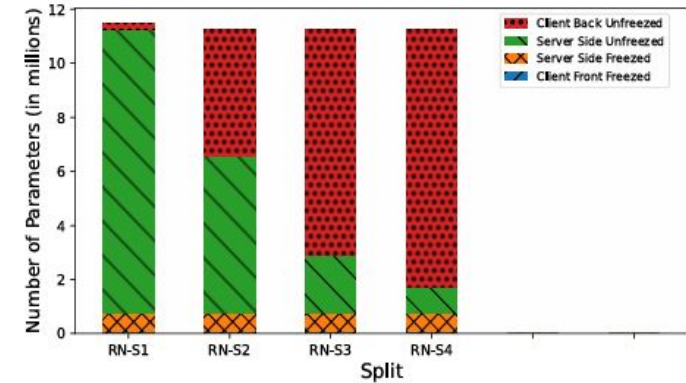


Figure: Statistics of various splits: Resnet-18.
The client front freeze is at the bottom of each bar, with 5k parameters and therefore not visible

Observation:

- When we are dealing with iid data across clients, how we split the model and number of trainable layers at client side does not matter.

Conclusion:

- We can offload maximum on server side.

Non-IID: Different Model Splits Results

small →

Large →

Model Split	Number of Parameters at Client Back (in million)	Generalised Model			Personalised Model		
		Test Acc.	Std	Epoch	Test Acc.	Std	Epoch
RN-S1	0.005	82.3	4.21	29	86.65	2.43	25
RN-S2	4.7	82.2	3.15	25	87.51	2.48	23
RN-S3	8.3	82.1	3.12	26	88.55	2.37	21
RN-S4	9.5	82.4	3.11	27	89.33	1.52	18

Performance metrics (Average Test Accuracy, Standard Deviation, and Epoch) for different splits in a Non-IID setting for 500 data-points on CIFAR-10 dataset with 10 clients

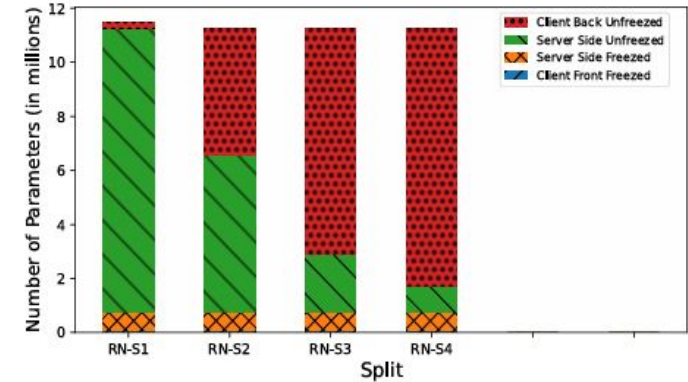


Figure: Statistics of various splits: Resnet-18.
The client front freeze is at the bottom of each bar, with 5k parameters and therefore not visible

Observations:

- When we are dealing with non- iid data across clients, how we split the model and number of trainable layers at client side matters.

Conclusion:

- We have to be careful when selecting the split.

Non-IID: Customization of Client Back Model

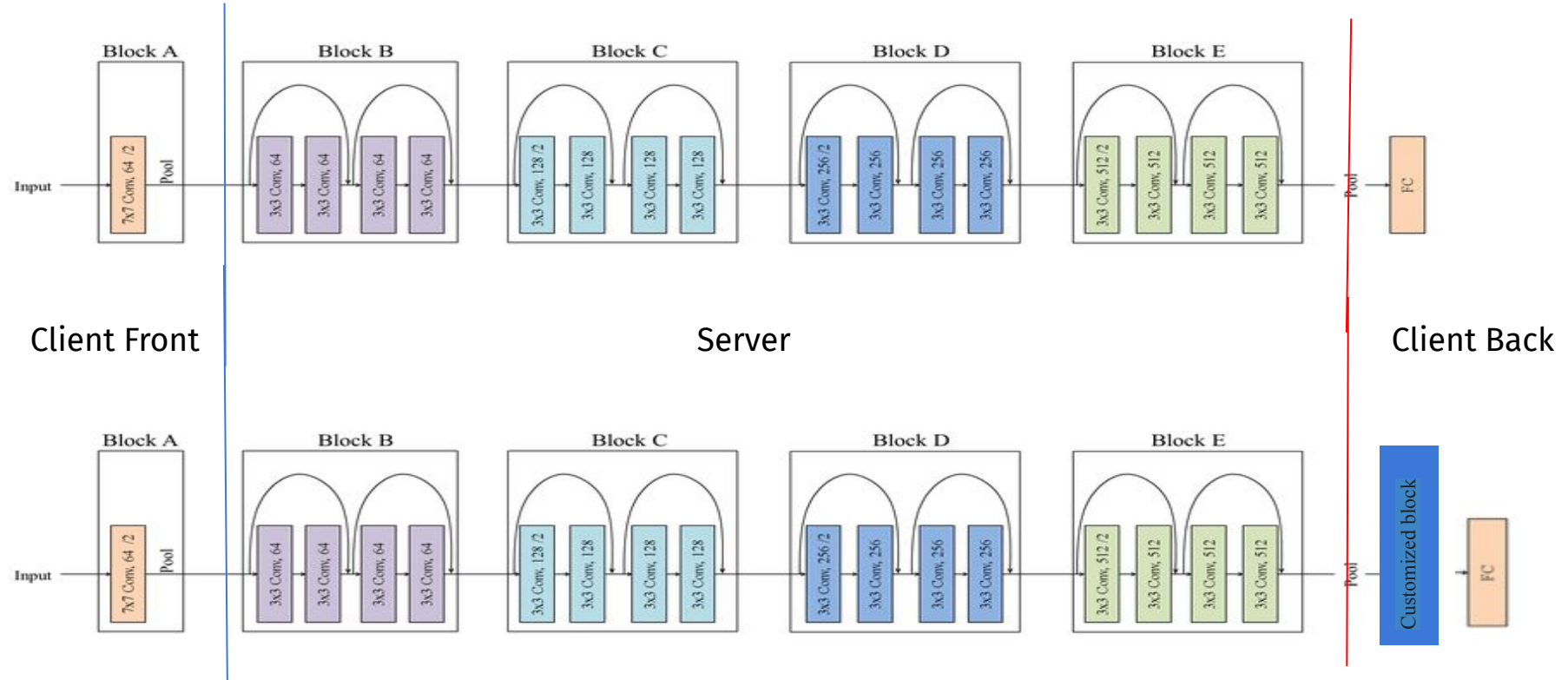
- There is a trade-off between resource requirements and performance in split learning when deciding how to allocate the model's parameters between the client and server.
- Increasing the parameters on the client's side allows for more personalized and fine-grained learning, leading to improved performance and accuracy.

Question:

- Can we customize the model and get the same performance with maximum offload on the server side ?

2. Effect of client back model customization.

Setup: Custom block



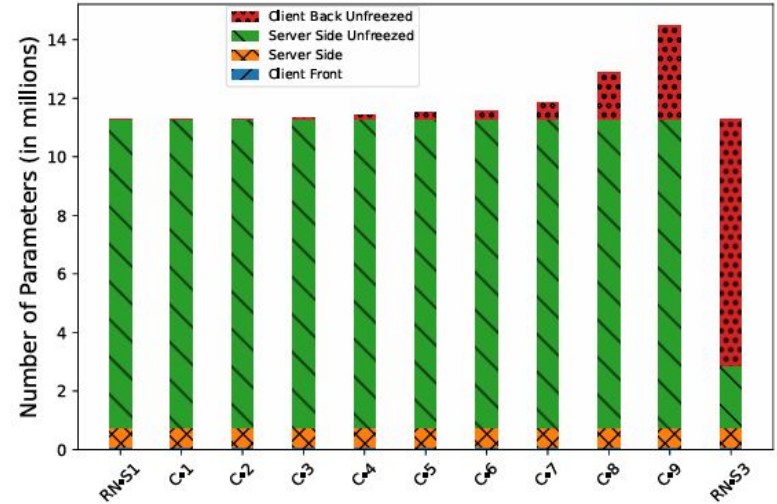
Result: Custom block

small →

Split	Client Back Side Model	No. of parameters (in million)	CIFAR-10	
			ESL-G	ESL-P
S-1	Linear Layer(512x10)	0.005	82.39	86.65
C-1	Linear Layer(512x64) Linear Layer(64x10)	0.033	81.65	87.75
C-2	Depthwise-Seperable(64x5x5) Linear Layer(64x10)	0.039	82.9	86.4
C-3	Depthwise-Seperable(128x5x5) Linear Layer(128x64) Linear Layer(64x10)	0.08	83.1	86
C-4	Depthwise-Seperable(256x5x5) Linear Layer(256x64) Linear Layer(64x10)	0.15	83.1	86.9
C-5	Depthwise-Seperable(512x5x5) Linear Layer(512x10)	0.28	<u>84.02</u>	<u>88.36</u>
C-6	Convolution Layer(64x5x5) Linear Layer(64x10)	0.29	82.6	86.8
C-7	Convolution(128x5x5) Linear Layer(128x64) Linear Layer(64x10)	0.59	81.8	87.3
C-8	Convolution(64x3x3) Linear Layer(64x10)	1.6	84.00	86.3
C-9	Convolution(128x3x3) Linear Layer(128x64) Linear Layer(64x10)	3.22	83.62	86.30
S-3	Last Convolution block and Linear Layer(512x10) of Resnet-18 model	8.39	<u>82.34</u>	<u>88.52</u>

Large →

Comparison of performance metric for different configurations of Client Back Models across various splits on CIFAR-10 dataset with 10 clients. The best result is underlined.



Statistics of various Customized splits: Resnet-18

The client front freeze is at the bottom of each bar, with 5k parameters and therefore not visible.

We've significantly improved ESL-G and achieved ESL-P at a comparable level, all while reducing the number of parameters in the client's backend model by **300X** times..

Effect of client back model customization:

	Split	Client Back Side	Number of parameters	CIFAR-10		FMNIST	
				ESL-G	ESL-P	ESL-G	ESL-P
small →	RN-S1	LL(512x10)	0.005	82.39	86.65	82.46	<u>91.48</u>
	C-1	2LL(512x64X10)	0.033	81.65	87.75	82.02	89.72
	C-5	DS(512x5x5)+LL(512x10)	0.28	<u>84.02</u>	88.36	<u>84.47</u>	90.53
Large →	RN-S3	Last CB+LL(512x10)	8.39	82.34	<u>88.52</u>	82.55	88.97

Comparison of performance metric for different configurations of Client Back Models across various splits on various datasets.
The best result is underlined.

Observation:

- By carefully selecting and designing the custom block, it is possible to achieve comparable or even improved performance compared to a larger, non-customized model. This allows for a higher offload of parameters to the server side while maintaining good accuracy and performance.

Conclusion: Customization allows for a higher offload of parameters to the server side while maintaining good accuracy and performance.

3. Overall Performance comparison with different algorithms.

Algorithms for comparative study:

- **ESL-G and ESL-P** : Efficient Split Learning Generalised and Personalised Model
- **ESL-G^o and ESL-P^o** : Personalized & Fair Split Learning Generalised and Personalised Model
- **FedAvg** : Federated Learning [1]
- **Fedavg_TL** : Federated Learning with Transfer Learning
- **SL** : Vanilla Split learning
- **SFLv2** : SplitFedv2 (Split Learning +federated Learning)
- **FedProx_TL** : Improved FedAvg [2]
- **Scaffold_TL** : Improved FedAvg [3]
- **UN-G and UN-P**: ResNet-18 pretrained weights with all the layers unfrozen for generalization and personalization in training.
- **NF-G and NF-P**: ResNet-18 pretrained weights with adjustments made to the last layer based on the dataset requirements.

Overall Performance Comparison-CIFAR10

Motivation: Standard Dataset

Set Up:

- Total 10 clients, Non-iid data distribution.
- Train(500 samples) and Test(1000 samples) data distribution is same for each client.
- Benefits of personalization and model customization is quantified.

Observation and Conclusion:

- ESL's performance aligns closely with Pooled emphasizes the effectiveness of our method in handling decentralized or Non-IID scenarios.
- ESL demonstrating **30%** and **35%** better performance over FedAvg_TL and SLv2, respectively.

Method	CIFAR-10	FMNIST	ISIC	DR
ESL-G	84.02	84.47	61.13	53.46
ESL-P	88.36	91.53	71.40	61.30
Pooled_TL	89.12	91.88	75.50	65.4
ESL-G ^o	82.39	82.46	61.90	50.87
ESL-P ^o	86.65	91.48	68.97	56.50
NF-G	70.25	68.57	49.21	41.74
NF-P	78.97	83.65	59.82	46.51
UN-G	85.15	85.57	55.30	46.56
UN-P	87.11	91.95	60.11	49.32
FedAvg	61.01	72.42	48.95	40.21
SL	65.42	76.22	51.95	41.45
SLV2	65.23	76.80	49.78	41.25
FedAvg_TL	67.60	78.71	54.11	42.05
FedProx_TL	67.06	72.51	55.22	42.06
Scaffold_TL	72.01	77.33	54.82	46.42

Performance comparison across different methods on various datasets with Non-IID setting.

Convergence Curve

Observation and Conclusion:

- ESL have faster and smoother convergence than competing techniques.
- When personalization begins, all loss curve slopes increase significantly, indicating accelerated convergence.
- Unfreezing all layers (UN-G and UN-P) has result in a slight accuracy increase for CIFAR-10.
- NF-G and NF-P exhibit lower accuracy compared to ESL-G and ESL-P, indicating the limitations of solely relying on pre-training without fine-tuning.

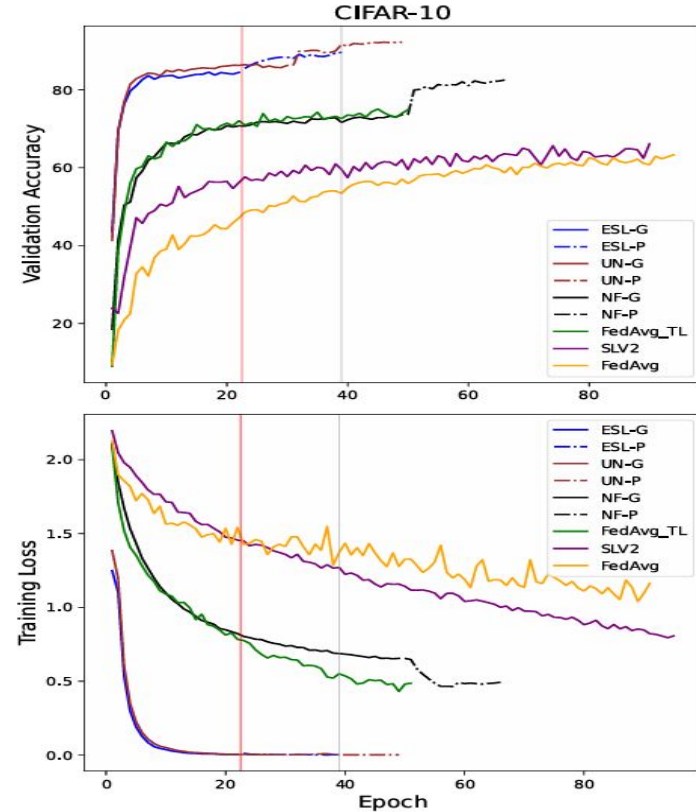


Fig: The convergence curves for training loss, and validation performance for CIFAR-10 using various methodologies.

Overall Performance Comparison-FMNIST

Motivation: Standard Dataset

Set Up:

- Total 10 clients, Non-iid data distribution.
- Train(750 samples) and Test(1000 samples) data distribution is same for each client.
- Benefits of personalization and model customization is quantified.

Observation and Conclusion:

- ESL's performance aligns closely with Pooled emphasizes the effectiveness of our method in handling decentralized or Non-IID scenarios.
- ESL demonstrating **16%** and **19%** better performance over FedAvg_TL and SLv2, respectively.

Method	CIFAR-10	FMNIST	ISIC	DR
ESL-G	84.02	84.47	61.13	53.46
ESL-P	88.36	91.53	71.40	61.30
Pooled_TL	89.12	91.88	75.50	65.4
ESL-G ^o	82.39	82.46	61.90	50.87
ESL-P ^o	86.65	91.48	68.97	56.50
NF-G	70.25	68.57	49.21	41.74
NF-P	78.97	83.65	59.82	46.51
UN-G	85.15	85.57	55.30	46.56
UN-P	87.11	91.95	60.11	49.32
FedAvg	61.01	72.42	48.95	40.21
SL	65.42	76.22	51.95	41.45
SLv2	65.23	76.80	49.78	41.25
FedAvg_TL	67.60	78.71	54.11	42.05
FedProx_TL	67.06	72.51	55.22	42.06
Scaffold_TL	72.01	77.33	54.82	46.42

Performance comparison across different methods on various datasets with Non-IID setting.

4. Evaluation of key-value store: Communication Reduction

Communication Reduction:

Method	Epochs till convergence	Client Front to Server (Activation)	Key in KV Store	Server to/from Client Back (Activation+gradient)	Model Weights to Server	Total data-transfer till convergence	Reduction compared to other method
ESL-G	22	401,408,000	22,000	45,056,000	902,880	447,388,880	—
ESL-P	20	0	0	0	0	0	—
Total ESL	42	401,408,000	22,000	45,056,000	902,880	447,388,880	—
ESL-G ⁻	22	8,830,976,000	0	45,056,000	902,880	8,876,934,880	—
ESL-P ⁻	20	8,028,160,000	0	20,480,000	0	8,048,640,000	—
Total ESL ⁻	42	16,859,136,000	0	65,536,000	902,880	16,925,574,880	37.8X
UN-G	30	24084480000	0	61440000	2288640	24148208640	—
UN-P	18	14450688000	0	36864000	0	14487552000	—
Total UN	48	38535168000	0	98304000	2288640	38635760640	86.35X
FL	94	0	0	0	8,408,594,784	8,408,594,784	18.8X
FL_TL	50	0	0	0	3,359,543,200	3,359,543,200	7.5X
SL	92	72,253,440,000	0	188,416,000	3,775,680	72,437,760,000	161.9X

Communication overhead (in Bytes) of various methods on CIFAR-10 dataset with Non-IID setting per client.

ESL⁻ is without Key-Value Store version of ESL

Observations: Upto **7.5X** and **161.9X** reduction in communication overhead per Client as compared to FL_TL and SL, during training.

Conclusion: The key-value store approach is even better with personalisation.

Comparison: ESL Vs Sparsification Technique

Paper: “Reducing Communication for Split Learning by Randomized Top-k Sparsification[11]”.

Dataset Used: CIFAR-100[36]

Model: ResNet-18

Observation:

- A 13% better accuracy with reduced communication overhead during the training phase.

Method	Accuracy (%)	Compressed Size
TOP-K Sparsification	66.01	12.5
ESL-P	75.1	1.21

Comparison of Accuracy and Compressed Size of ESL with [11]. We assume the original communication size(non-compression case) to be 100.

5. Evaluation of key-value store: Computation Reduction

Computation Reduction:

Method	Total Epoch till convergence	Client Side Front Layer	Server Side Central Layer	Client Side Back Layer	Total GLOPS till convergence	Reduction compared to other method
ESL-G	22	0.242	0	0.000676	0.243126	—
ESL-P	20	0	0	0.000614	0.000614	—
Total ESL	42	0.242	0	0.001290	0.243740	—
ESL-G ⁻	22	5.333	0	0.000676	5.334585	—
ESL-P ⁻	20	4.849	0	0.000614	4.849623	—
Total ESL ⁻	42	10.18	0	0.001290	10.18421	41.78X
Total UN	48	21.82	0	0.001474	34.91433	143.24X
FL	94	68.37	960.1	0.002887	1028.469	4219.5X
FL.TL	50	12.12	334.9	0.001536	347.0689	1423.9X
SL	92	66.92	0	0.002826	66.91914	274.4X

Floating-point operations at the client in various methods on CIFAR-10 dataset with Non-IID setting per client. Without the Key-Value Store version of ESL is shown as ESL⁻.

Observations: Upto 4216X reduction in GFLOPs/Work done per Client, during training.

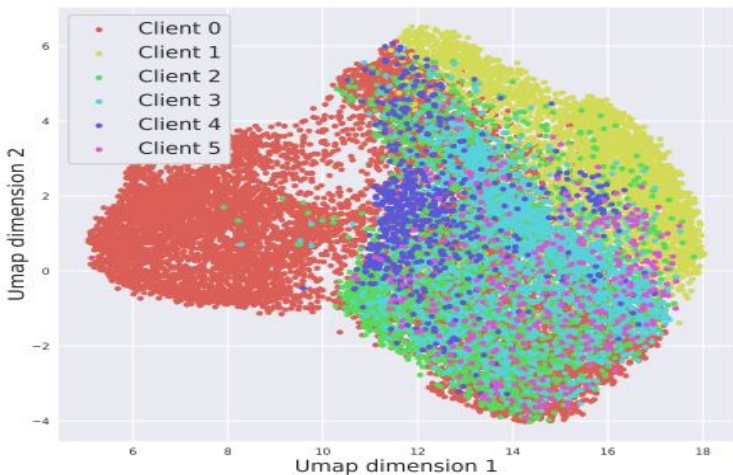
Conclusion: The key-value store approach is even better with personalisation.

6. Skin Lesion Diagnosis

Fed-ISIC2019 Dataset

Motivation:

- Natural Splits(HealthCare-Skin Disease)
- High-quality dermoscopy images of skin lesions
- Limited Clients(Cross-Silo Setting)
- Real-world problems:
 - heterogeneity between clients
 - high label imbalance
 - Multiclass classification



UMAP of deep network features of the raw images, colored by clients.

Main Challenge:

Model training bias or poor performance on certain clients.

Number	Client	Dataset size	Train	Test
0	Hospital Clínic de Barcelona	12413	9930	2483
1	ViDIR Group, Medical University of Vienna (MoleMax HD)	3954	3163	791
2	ViDIR Group, Medical University of Vienna (DermLite FOTO)	3363	2691	672
3	The skin cancer practice of Cliff Rosendahl	2259	1807	452
4	Memorial Sloan Kettering Cancer Center	819	655	164
5	ViDIR Group, Medical University of Vienna (Heine Dermaphot)	439	351	88

Information for the different clients in Fed-ISIC2019

Fed-ISIC2019 Dataset

1. Metric Used: [Balanced Accuracy](#)[27]

(The average of the recalls calculated for each class)

It accounts for both the positive and negative outcome classes and doesn't mislead with imbalanced data.

2. Loss Function Used: [Weighted Focal Loss](#)[35]

Focal loss applies a modulating term to the cross entropy loss in order to focus learning on hard misclassified examples.

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t)$$

p_t : probability output by our model for the ground-truth class

α_t : is the weight of the ground-truth class

γ : is a hyperparameter

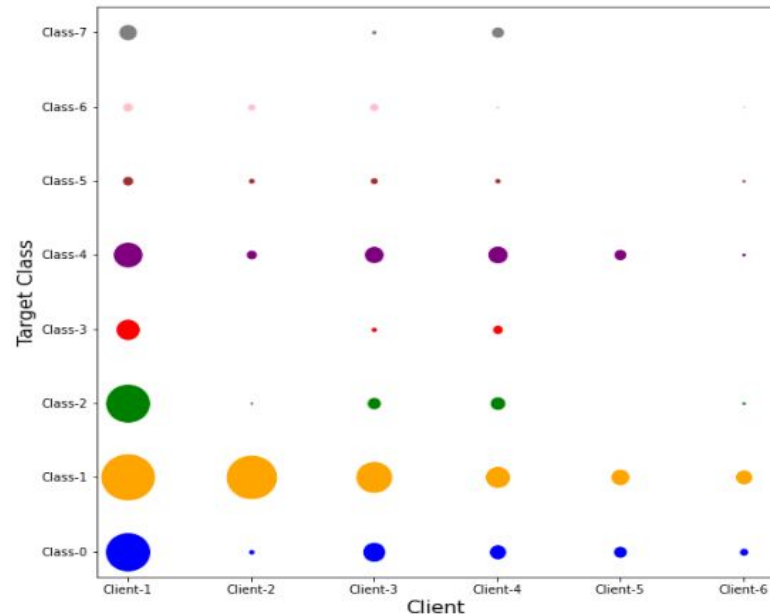


Figure: Data Distribution among clients ,where larger circle indicates more number of training samples for that particular class and client

Fed-ISIC2019 Dataset: ESL Vs Flamby[23]

Observation and Conclusion:

- ESL-P demonstrating **5.7%** better performance over FedAvg.
- ESL-P demonstrating **22%** better performance over FedAvg.
- It shows lower variability(standard deviation).

Method	Balanced Accuracy(%)	Standard Deviation
FedAvg	58.06	13.19
FedProx	55.02	13.21
Scaffold	61.9	13.90
ESL-G	61.1	13.18
ESL-P	71.4	11.62

Performance comparison across different methods on Fed-ISIC2019 dataset.

Overall Performance Comparison-ISIC2019

Observation and Conclusion:

- ESL's performance aligns closely with Pooled emphasizes the effectiveness of our method in handling decentralized or Non-IID scenarios.
- ESL demonstrating **32%** and **45%** better performance over FedAvg_TL and SLv2, respectively.

Method	CIFAR-10	FMNIST	ISIC	DR
ESL-G	84.02	84.47	61.13	53.46
ESL-P	88.36	91.53	71.40	61.30
Pooled_TL	89.12	91.88	75.50	65.4
ESL-G ^o	82.39	82.46	61.90	50.87
ESL-P ^o	86.65	91.48	68.97	56.50
NF-G	70.25	68.57	49.21	41.74
NF-P	78.97	83.65	59.82	46.51
UN-G	85.15	85.57	55.30	46.56
UN-P	87.11	91.95	60.11	49.32
FedAvg	61.01	72.42	48.95	40.21
SL	65.42	76.22	51.95	41.45
SLV2	65.23	76.80	49.78	41.25
FedAvg_TL	67.60	78.71	54.11	42.05
FedProx_TL	67.06	72.51	55.22	42.06
Scaffold_TL	72.01	77.33	54.82	46.42

Performance comparison across different methods on various datasets with Non-IID setting.

Convergence Curve

Observation and Conclusion:

- ESL have faster and smoother convergence than competing techniques.
- When personalization begins, all loss curve slopes increase significantly, indicating accelerated convergence.
- Unfreezing all layers (UN-G and UN-P) approach tends to lead to overfitting when applied to datasets from different domains, as seen with ISIC-2019.
- NF-G and NF-P exhibit lower accuracy compared to ESL-G and ESL-P, indicating the limitations of solely relying on pre-training without fine-tuning.

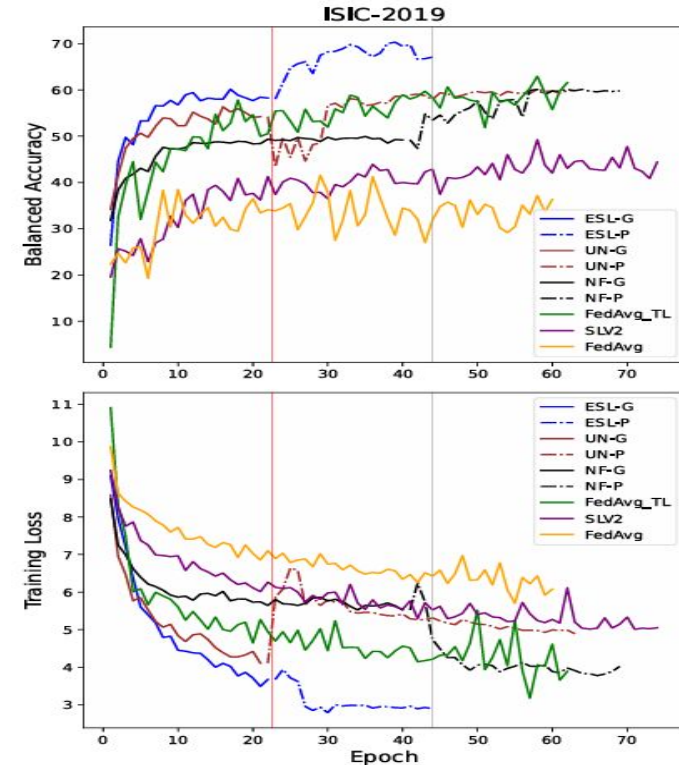


Fig: The convergence curves for training loss, and validation performance for ISIC-2019 using various methodologies.

7. Diabetic Retinopathy Detection

Overall Performance Comparison-Diabetic Retinopathy

Motivation: Distinct geographical locations and healthcare setting

Set Up:

- High-Resolution Retina Images
- Two datasets are used: EyePACS and APTOS.
- Total 10 clients, first 5 clients are provided the APTOS dataset and the next 5 clients are provided the EyePACS dataset.

Observation and Conclusion:

- ESL's performance aligns closely with Pooled emphasizes the effectiveness of our method in handling decentralized or Non-IID scenarios.
- ESL demonstrating **46%** and **49%** better performance over FedAvg_TL and SLv2, respectively.

Method	CIFAR-10	FMNIST	ISIC	DR
ESL-G	84.02	84.47	61.13	53.46
ESL-P	88.36	91.53	71.40	61.30
Pooled_TL	89.12	91.88	75.50	65.4
ESL-G ^o	82.39	82.46	61.90	50.87
ESL-P ^o	86.65	91.48	68.97	56.50
NF-G	70.25	68.57	49.21	41.74
NF-P	78.97	83.65	59.82	46.51
UN-G	85.15	85.57	55.30	46.56
UN-P	87.11	91.95	60.11	49.32
FedAvg	61.01	72.42	48.95	40.21
SL	65.42	76.22	51.95	41.45
SLV2	65.23	76.80	49.78	41.25
FedAvg_TL	67.60	78.71	54.11	42.05
FedProx_TL	67.06	72.51	55.22	42.06
Scaffold_TL	72.01	77.33	54.82	46.42

Performance comparison across different methods on various datasets with Non-IID setting.

Overall Performance Comparison across all datasets

Observation and Conclusion:

- On average, ESL demonstrating 31% and 37% better performance over FedAvg_TL and SLv2, respectively.
- ESL's performance aligns closely with Pooled or centralised training performance which emphasizes the effectiveness of our method in handling decentralized or Non-IID or class imbalance scenarios.

Method	CIFAR-10	FMNIST	ISIC	DR
ESL-G	84.02	84.47	61.13	53.46
ESL-P	88.36	91.53	71.40	61.30
Pooled_TL	89.12	91.88	75.50	65.4
ESL-G ^o	82.39	82.46	61.90	50.87
ESL-P ^o	86.65	91.48	68.97	56.50
NF-G	70.25	68.57	49.21	41.74
NF-P	78.97	83.65	59.82	46.51
UN-G	85.15	85.57	55.30	46.56
UN-P	87.11	91.95	60.11	49.32
FedAvg	61.01	72.42	48.95	40.21
SL	65.42	76.22	51.95	41.45
SLV2	65.23	76.80	49.78	41.25
FedAvg_TL	67.60	78.71	54.11	42.05
FedProx_TL	67.06	72.51	55.22	42.06
Scaffold_TL	72.01	77.33	54.82	46.42

Performance comparison across different methods on various datasets with Non-IID setting.

Evaluation of key-value store

Observations:

- From 1.16X to maximum 109X times reduction in communication overhead per Client.
- From 41X to maximum 4216X times reduction in GFLOPs/Work done per Client.

Conclusion:

- The key-value store approach in ESL provides a practical solution to reduce communication overhead and optimize computations.

Method	CIFAR-10	FMNIST	ISIC	DR
FL	18.8	8.21	1.94	2.94
FL_TL	7.51	5.21	1.16	1.80
SL	161.9	93.7	109.3	109.7
ESL ⁻	37.8	40.1	40.4	42.5

Reduction in Communication Overhead of ESL concerning different techniques across all datasets.

Without the Key-Value Store version of ESL is shown as ESL⁻

Method	CIFAR-10	FMNIST	ISIC	DR
FL	4219.5	2915.9	3499.6	3364.1
FL_TL	1423.9	1565.3	1765.0	1735.8
SL	274.55	164.03	181.97	184.88
ESL ⁻	41.783	46.727	44.751	47.715

Reduction in GFLOPs of ESL concerning different techniques across all datasets.

Without the Key-Value Store version of ESL is shown as ESL⁻

8. Brain Image Segmentation

Fed-IXI-Tiny Dataset

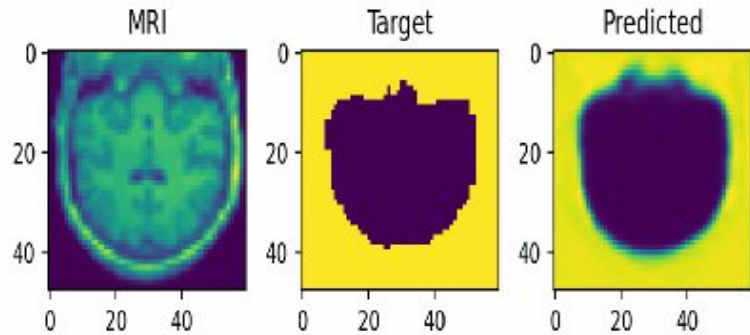
Motivation:

- T1-weighted brain MR images along with a set of corresponding brain image segmentations labels, taking the form of binary image masks.
- Natural Splits from three different London hospitals.
- Limited Clients(Cross-Silo Setting)

Task: The task is to segment the brain on the volume.

Hospital Name	Sex	Dataset size	Age
Guys	Female	184	53.23 ± 15.25
	Male	144	51.02 ± 17.26
HH	Female	93	50.28 ± 16.93
	Male	85	44.43 ± 15.67
IOP	Female	44	43.90 ± 18.43
	Male	24	39.57 ± 12.46

Demographics information for Fed-IXI.



Brain MRI images, Target Mask and Predicted Mask

Fed-IXI-Tiny Dataset

1. **Metric Used:** Dice Score[37]

It measures the overlap between the segmented region and ground truth, providing a comprehensive measure of segmentation accuracy.

2. **Loss Function Used:** Dice Loss[38]

The model was directly trained for the DICE loss

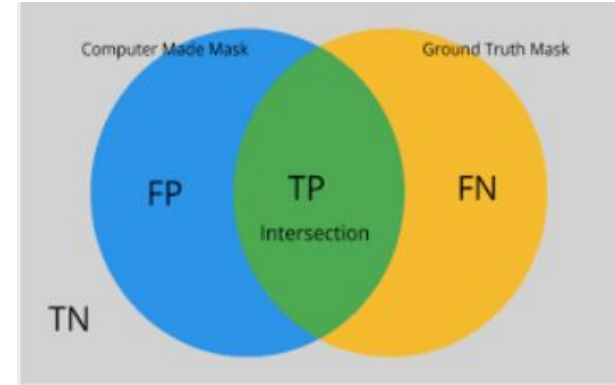
$$\ell_{DICE} = 1 - S_{DICE} = 1 - \frac{2TP}{2TP + FP + FN + \epsilon},$$

TP : True positive rate

FP : False positive rate,

FN : False negative rate

ϵ : Ensures numerical stability



Dice Score Visualization

Main Challenge - Where to split?

- Symmetric architecture with two main parts: the contracting path and the expansion path.
- Splitting at the last layers significantly communication because of larger size activations.
- Splitting in the middle layers significantly boosts client-side computations.
- Deciding where to split such a model is quite challenging.

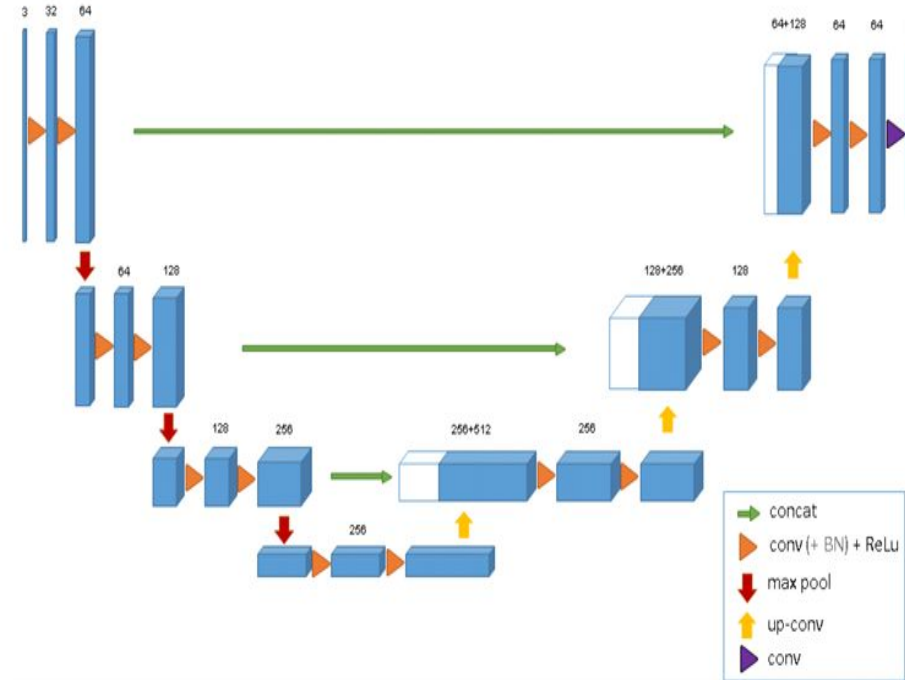


Fig: The 3D U-Net architecture. Blue boxes represent feature maps.
The number of channels is denoted above each feature map[32]

Setup Split-1

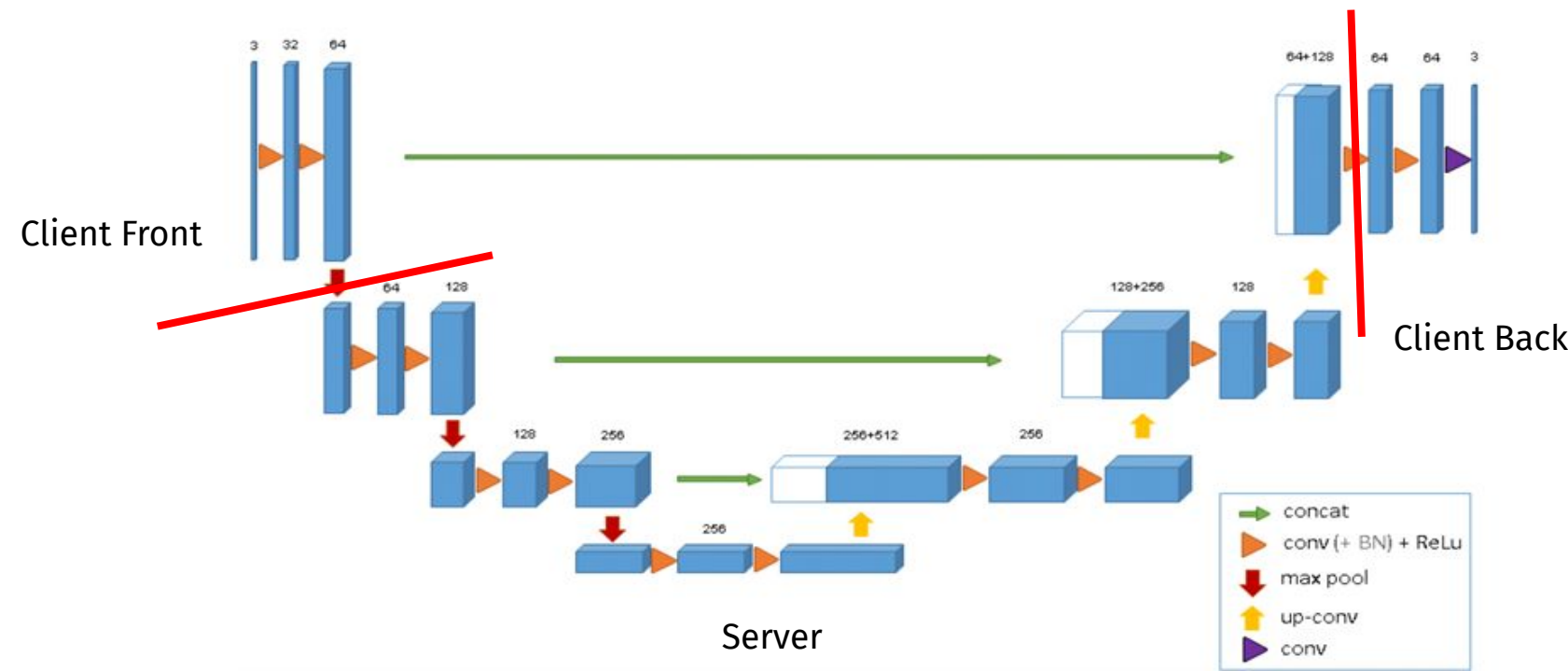


Fig: The 3D U-Net architecture. Blue boxes represent feature maps.
The number of channels is denoted above each feature map [32].

Setup Split-2

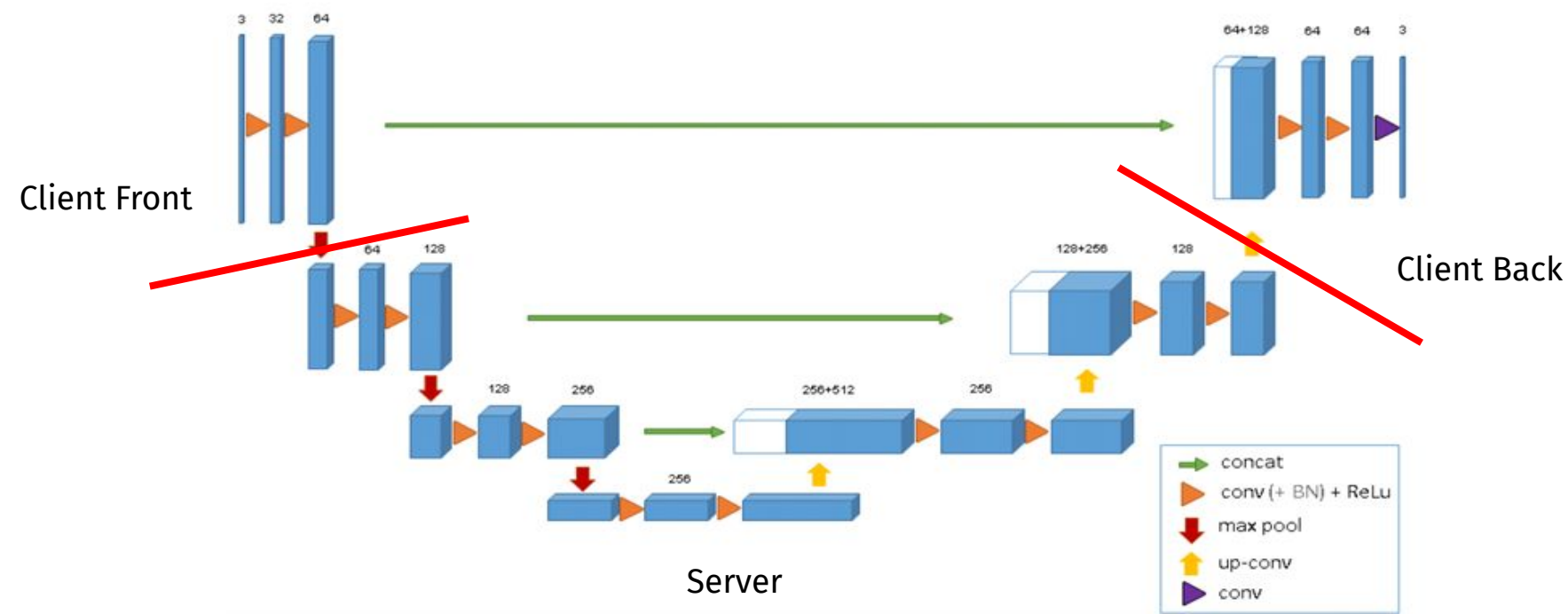


Fig: The 3D U-Net architecture. Blue boxes represent feature maps.
The number of channels is denoted above each feature map[32]

Setup Split-3

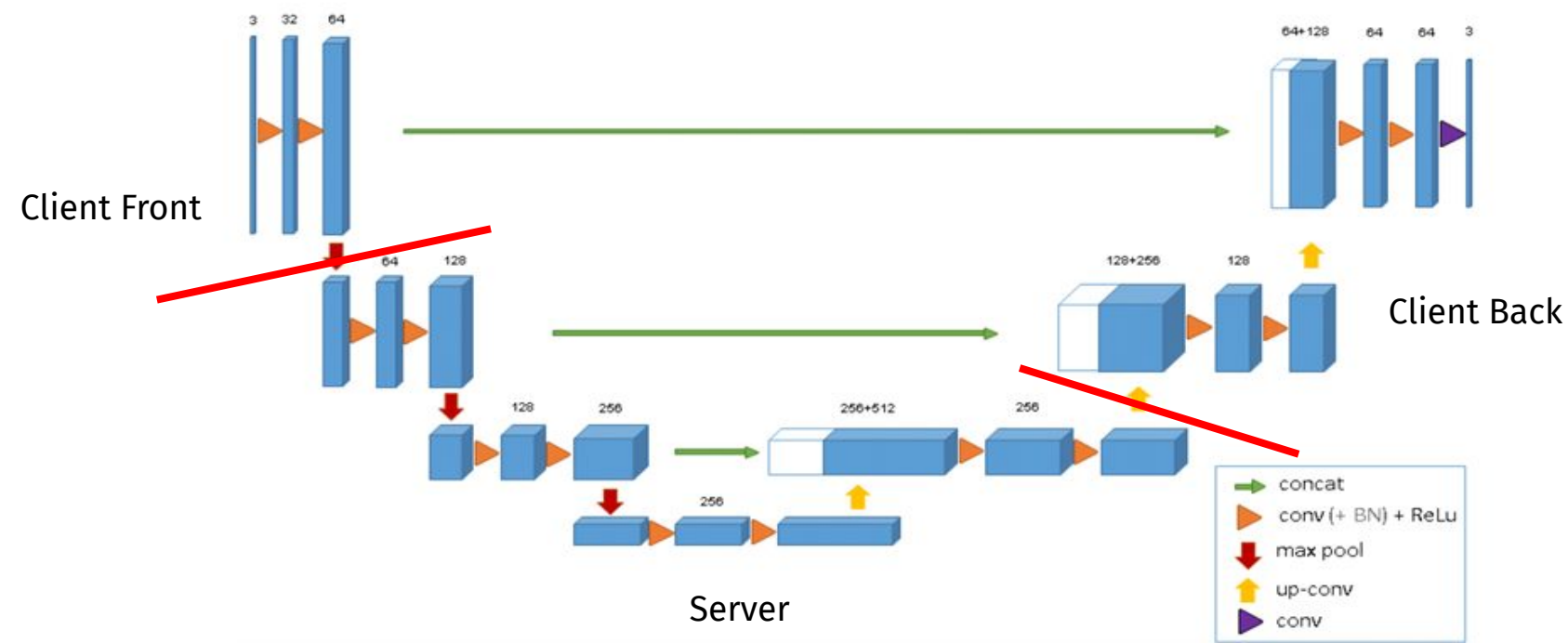


Fig: The 3D U-Net architecture. Blue boxes represent feature maps.
The number of channels is denoted above each feature map [32]

Setup Split-4

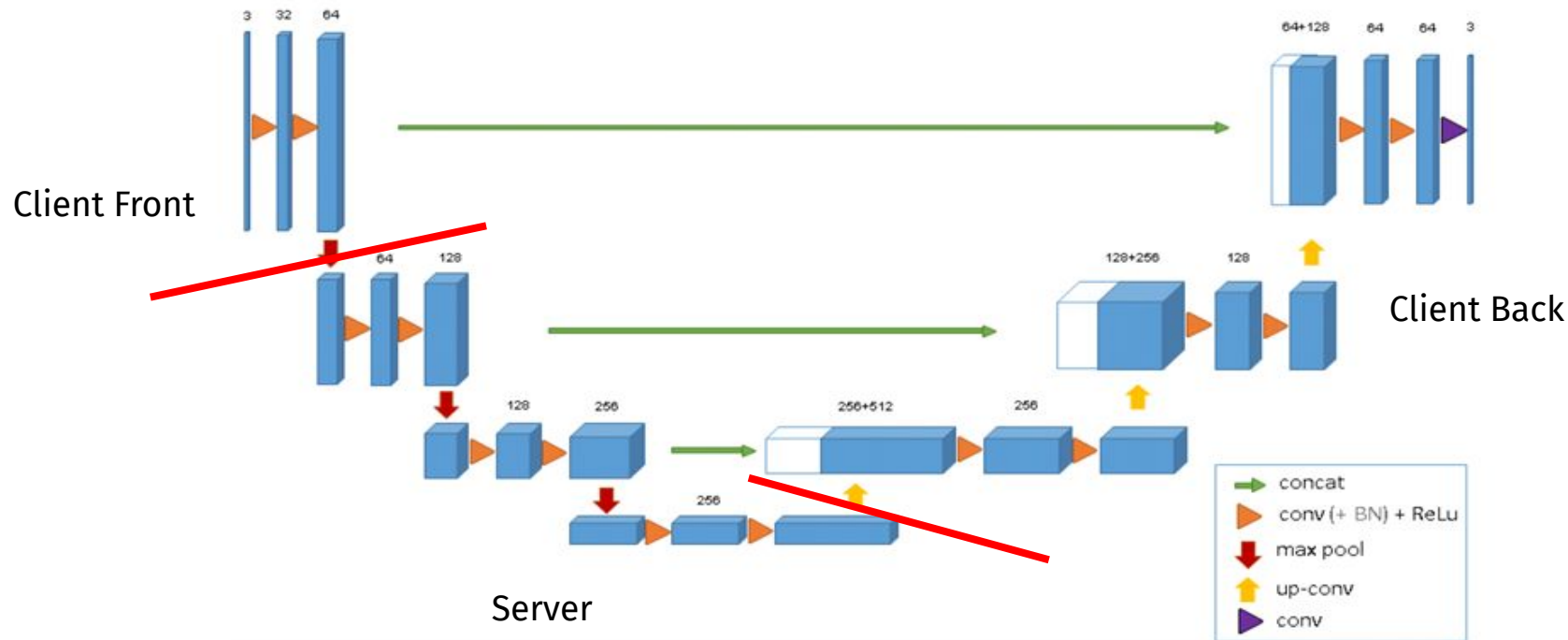


Fig: The 3D U-Net architecture. Blue boxes represent feature maps.
The number of channels is denoted above each feature map [32].

Performance Comparison for Different Splits

Model Split	Total data transfer till convergence	Total GFLOPs till convergence	Test Dice Dice score $\pm$ std dev	Epochs
UN-S1	7,941,624,851	4.60	0.782 ± 0.020	10
UN-S2	7,943,055,731	16.24	0.792 ± 0.001	10
UN-S3	2,547,619,481	22.02	0.877 ± 0.010	10
UN-S4	2,578,357,761	25.34	0.916 ± 0.001	10
UN-S5	926,034,663	22.02	0.872 ± 0.001	10
UN-S5*	926,034,663	25.02	0.876 ± 0.001	15

Performance metrics for different splits on IXI-Tiny dataset for 3D UNet model. Only global generalized model results except for UN-S5* where personalization after generalization has been performed

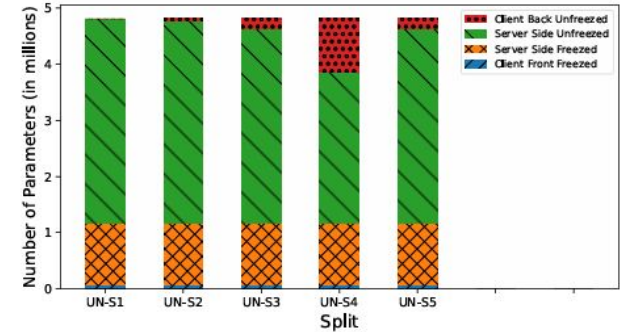


Figure: Statistics of various splits: 3D U-Net. The client front freeze is at the bottom of each bar, with 5k parameters and therefore not visible

Observations: Assigning more layers to the client's backside improved the Dice Score, but it is computationally demanding for the client. Also Personalization does not improve performance for 3D segmentation.

Conclusion: The optimal/best split depends on the resource constraints.

9. Kidney Tumor Segmentation

Fed-KITS-19 Dataset

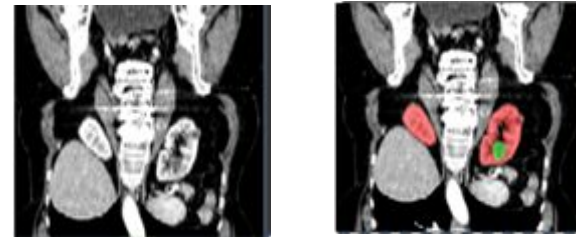
Motivation:

- The KiTS19 stems from the Kidney Tumor Segmentation Challenge 2019
- Natural Splits from six different London hospitals, contains CT scans of patients
- Limited Clients(Cross-Silo Setting)

Task: The task consists of both kidney and tumor segmentation, labeled 1 and 2, respectively.

Local ID Number	Dataset size	Train	Test
0	12	9	3
1	14	11	3
2	12	9	3
3	12	9	3
4	16	12	4
5	30	24	6

Information for the selected clients in Fed-KiTS19



An example of a coronal section of one of the training cases with its ground truth segmentation overlaid (kidney in red, tumor in blue).

Setup Split-1

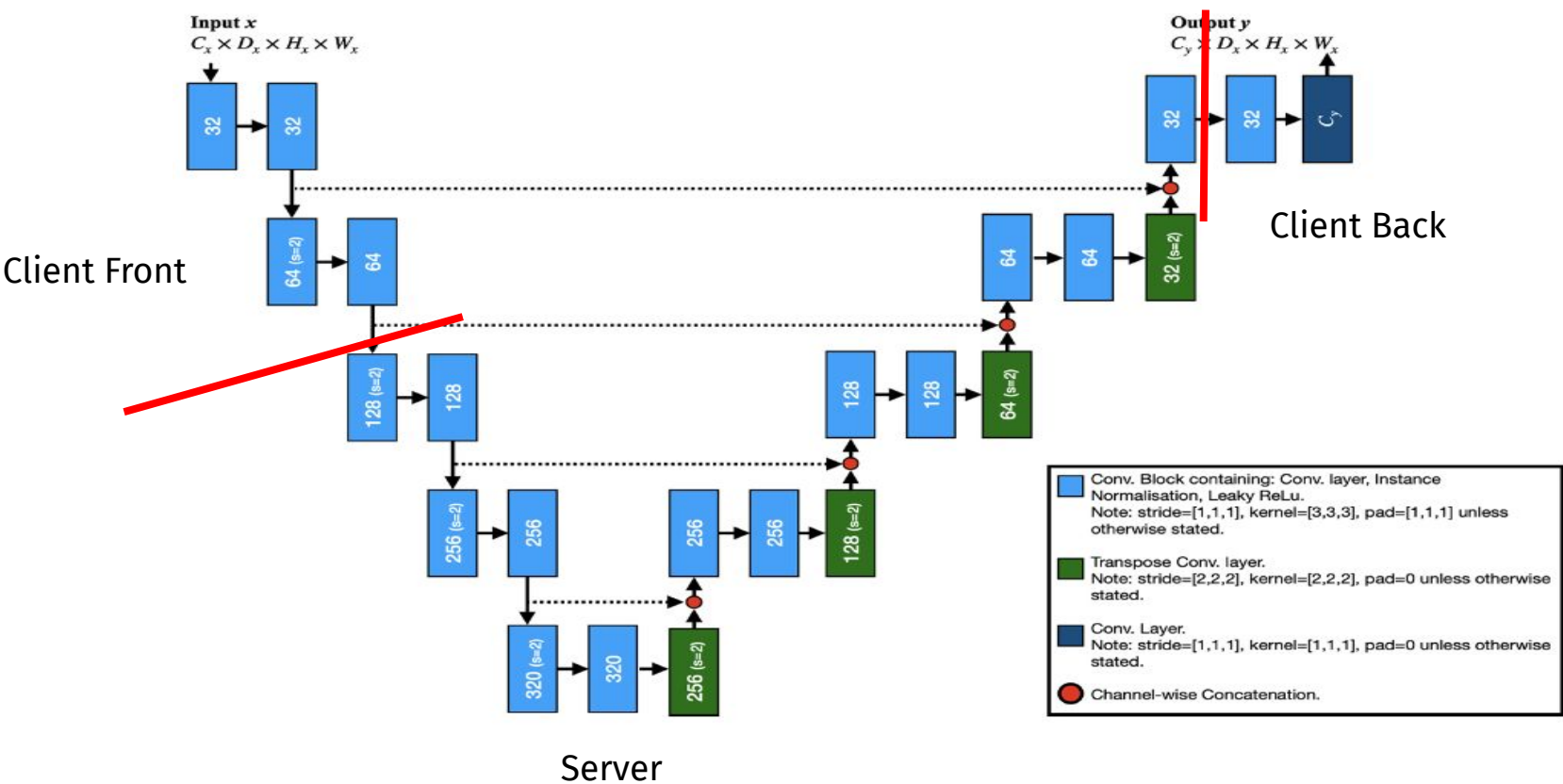


Figure: Baseline nnUNet architecture representation [36]

Setup Split-2

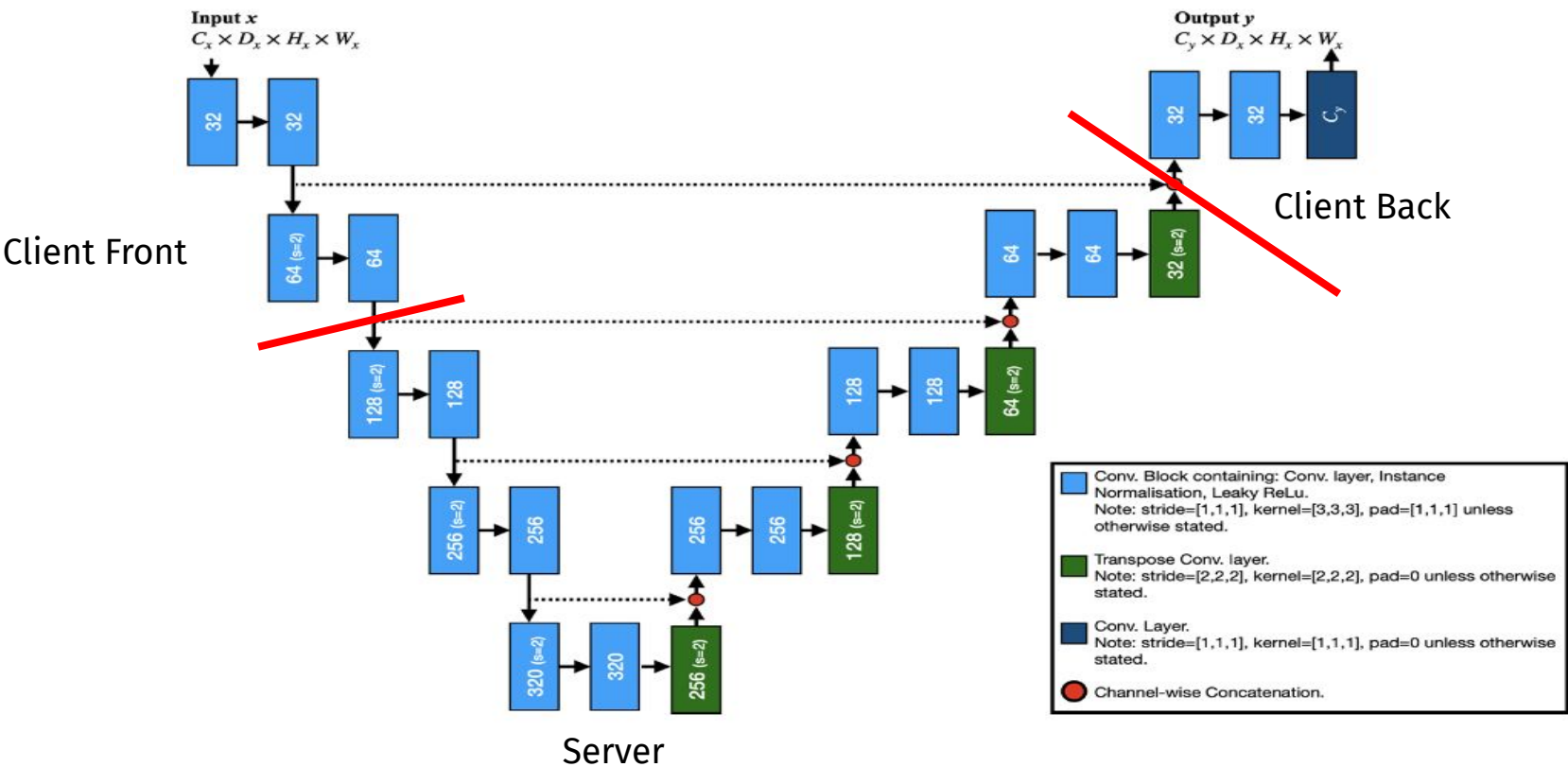


Figure: Baseline nnUNet architecture representation [36]

Setup Split-3

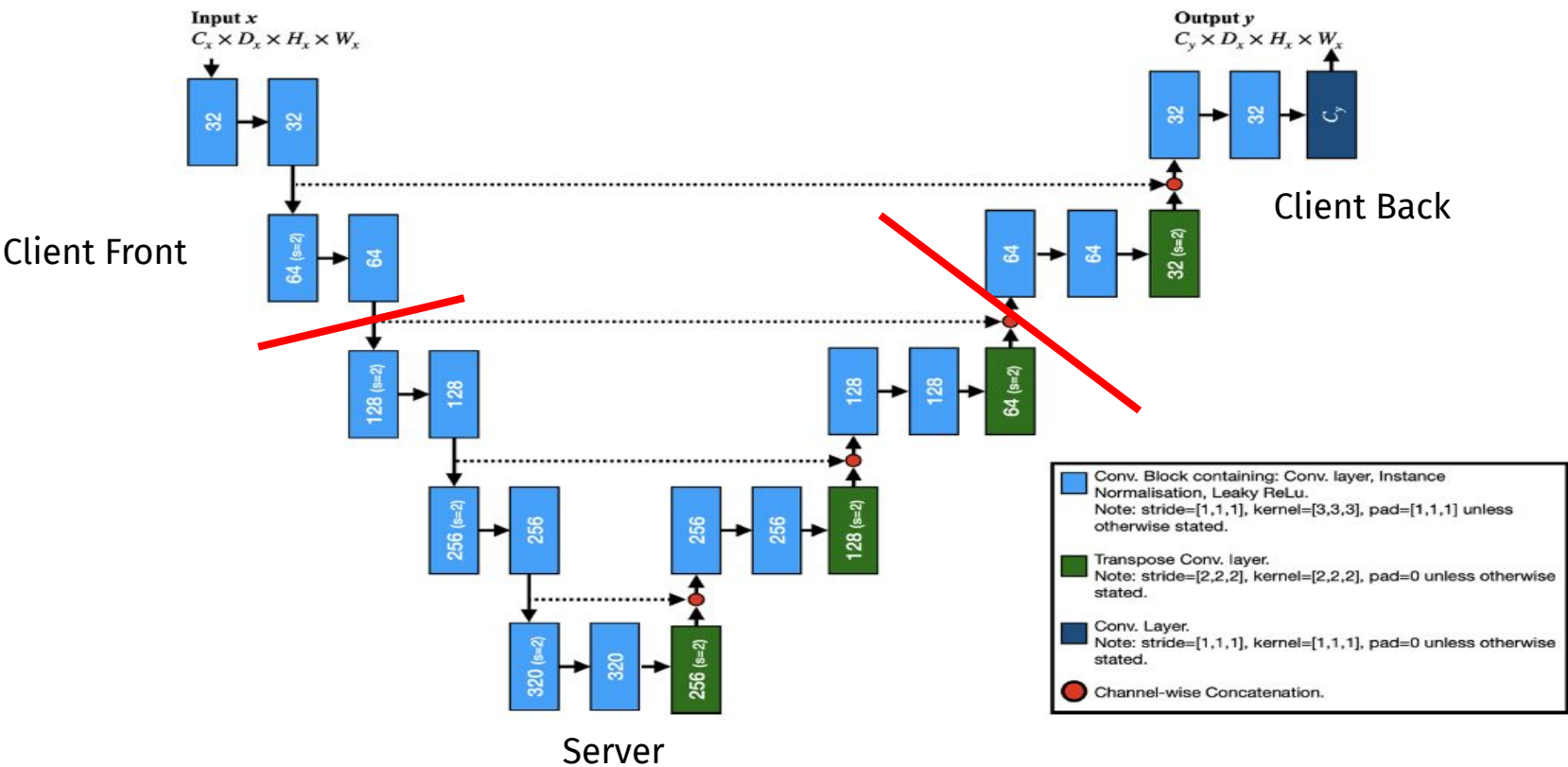


Figure: Baseline nnUNet architecture representation [36]

Setup Split-4

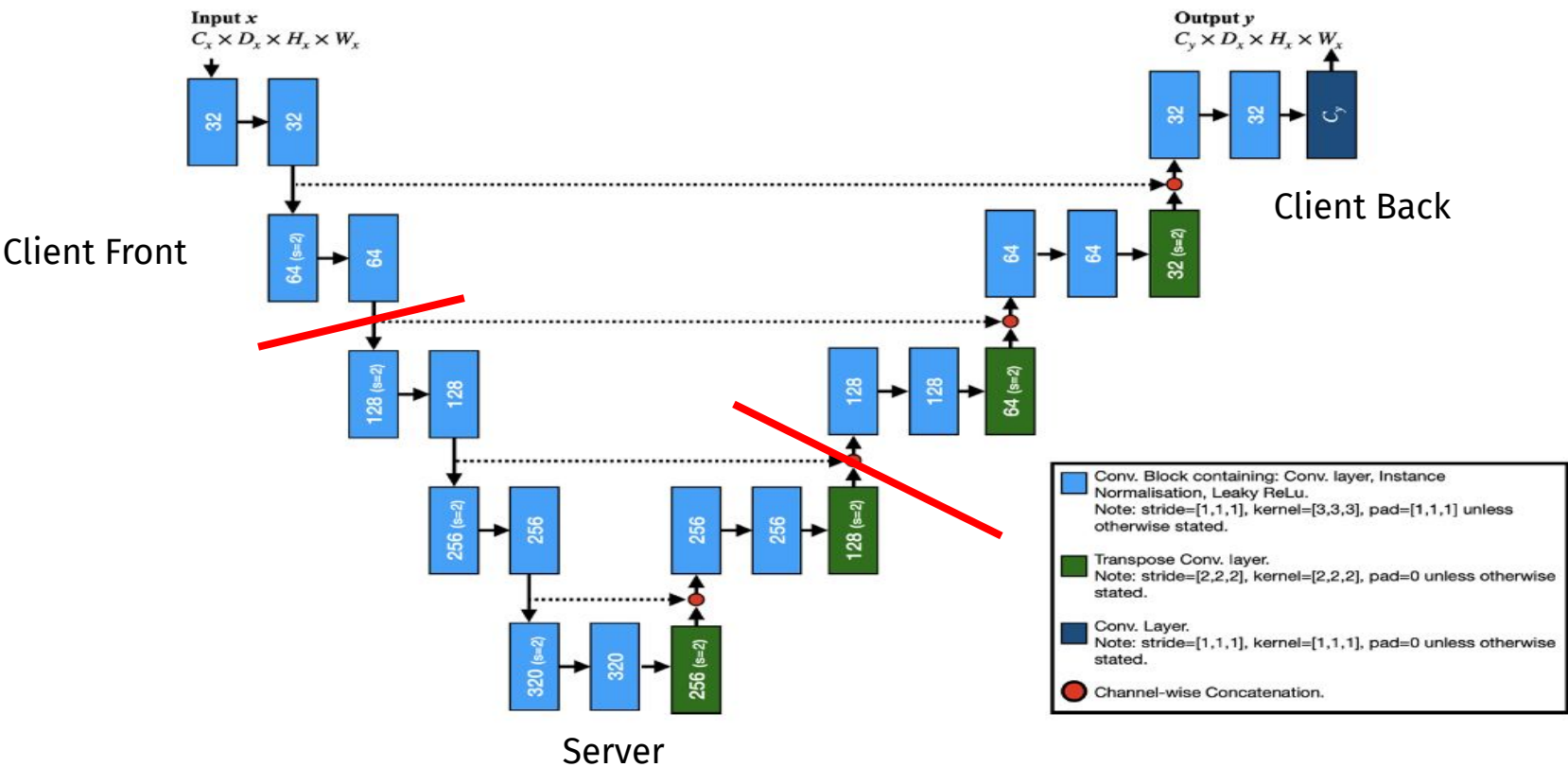


Figure: Baseline nnUNet architecture representation [36]

Setup Split-4

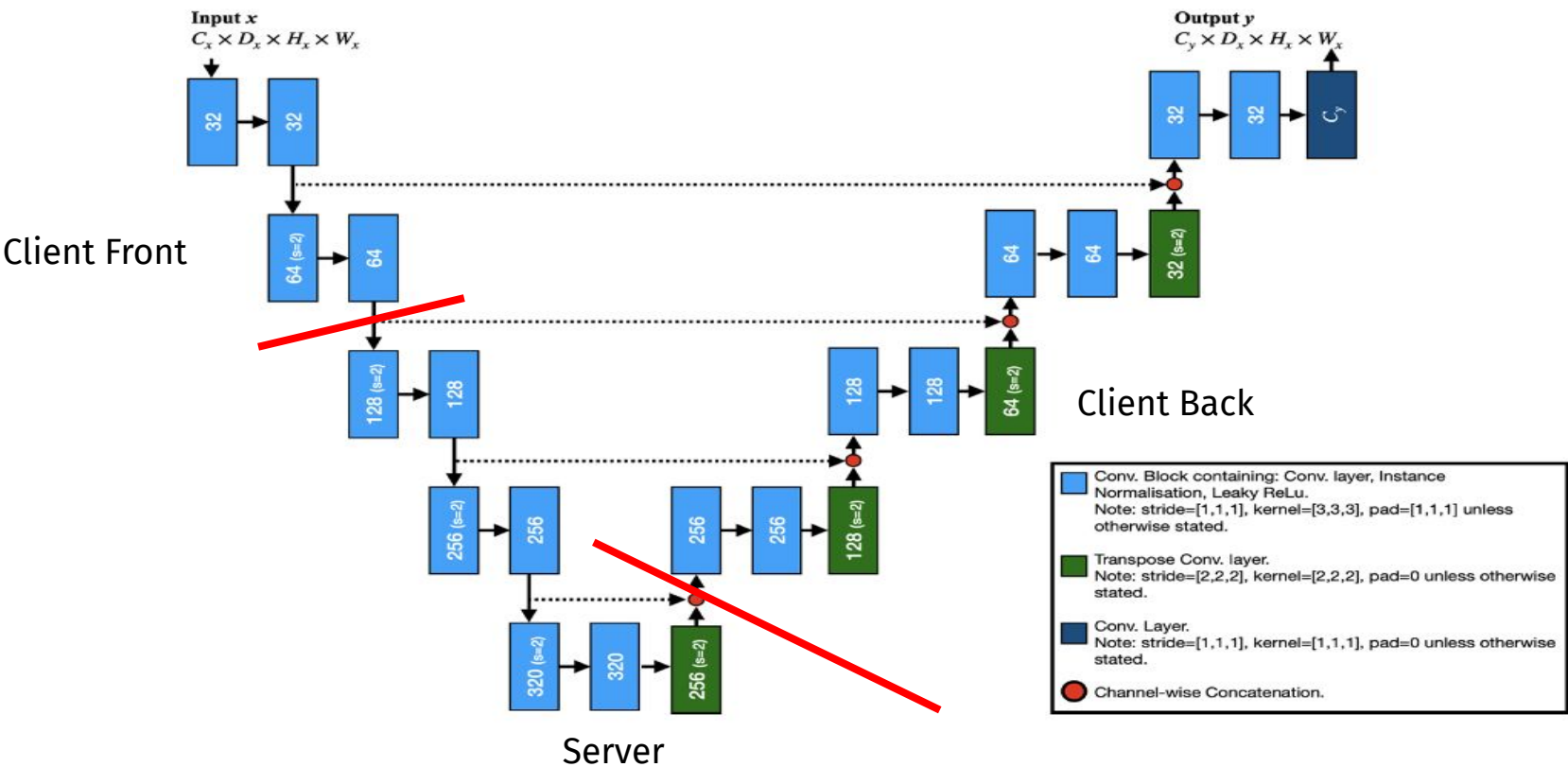


Figure: Baseline nnUNet architecture representation

Performance Comparison for Different Splits

Model Split	Total data transfer till convergence	Total GFLOPs till convergence	Test Dice Dice score $\pm$ std dev	Epochs
UN-S1	69,033,901,080	167,010.24	0.5655 $\pm$ 0.04	30
UN-S2	69,047,239,440	178,456.14	0.6747 $\pm$ 0.08	30
UN-S3	27,039,069,990	189,847.72	0.7463 $\pm$ 0.06	30
UN-S3*	27,039,069,990	195,662.02	0.7411 $\pm$ 0.08	30 + 5
UN-S4	14,479,142,910	237,309.65	0.8056 $\pm$ 0.03	30
UN-S5	15,113,392,140	239,682.74	0.8048 $\pm$ 0.02	30

Performance metrics for different splits on KITS-19 dataset for nnUNet model. Only global generalized model results except for UN-S3* where personalization after generalization has been performed

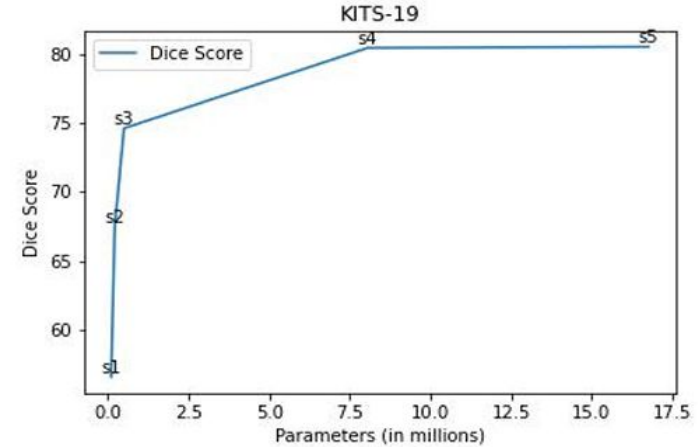


Figure: Parameters at client back side vs. dice Score for KITS-19 dataset for 3D nnU-Net model

Observations: Assigning more layers to the client's backside improved the Dice Score, but it is computationally demanding for the client. Personalization does not improve performance for 3D segmentation.

Conclusion: The optimal/best split depends on the resource constraints.

Overall Performance Comparison-3D Segmentation

Dataset	Technique	ESL	Pooled	FedAvg	FedProx	Scaffold	Communication Reduction	Computation Reduction
KITS-19	ESL(UN-S3)	0.746	0.626	0.514	0.561	0.553	2.38	45.43
IXI-Tiny	ESL(UN-S5)	0.872	0.954	0.769	0.768	0.769	0.20	2.34

Figure: Performance metrics for different splits on IXI-Tiny dataset for 3D UNet model.

Observation:

- KITS showed a **45%** improvement compared to FedAvg results.
- The IXI-Tiny dataset outperformed several algorithms, with an average improvement of **17%** over FedAvg results.

Conclusion

- Split learning is feasible and effective within federated domains.
- It is well-suited for edge computing applications, enabling on-device model training.
- Optimal performance can be achieved with customized models, reducing computational overhead.
- By careful configuration, we can reduce the communication and computation overhead of split learning without reduction model accuracy.
- Its applicability extends to various domains like NLP, where significant computational resources are typically required.

Future Work

- Support for Heterogeneous datasets and Heterogeneous Devices.
- Dynamic Model Selection and Customization.
- Support for out of order distributions.
- Secure Aggregation Techniques.
- Real-World Deployments.
- New architectures for Split Learning.

References



1. H.Brendan McMahan et. al. "Federated Learning of Deep Networks using Model Averaging". CoRR abs/1602.05629 (2016). arXiv: 1602.05629. URL: <http://arxiv.org/abs/1602.05629>.
2. Tian Li et. al. Federated Optimization in Heterogeneous Networks. 2020. arXiv: 1812. 06127 [cs.LG]
3. Sai Praneeth Karimireddy et. al. SCAFFOLD: Stochastic Controlled Averaging for Federated Learning. 2021. arXiv: 1910.06378 [cs.LG]
4. M. Yashwanth et. al. Federated Learning on Heterogeneous Data via Adaptive Self Distillation. 2023. arXiv: 2305.19600 [cs.LG]
5. Sebastian Caldas, Virginia Smith, and Ameet Talwalkar. "Federated kernelized multi task learning". Proc. SysML Conf. 2018, pp. 1–3
6. Y. Zhao, M. Li, L. Lai, N. Suda, D. Civin, and V. Chandra, "Federated learning with non-iid data," 2018. [Online]. Available: <https://arxiv.org/abs/1806.00582>
7. T. Yu, E. Bagdasaryan, and V. Shmatikov, "Salvaging federated learning by local adaptation," 2020. [Online]. Available: <https://arxiv.org/abs/2002.04758>
8. Yusuke Kanamori et. al. "An asynchronous federated learning focusing on updated models for decentralized systems with a practical framework". 2023 IEEE 47th Annual Computers, Software, and Applications Conference (COMPSAC). 2023, pp. 1147–1154. DOI: 10.1109/COMPSAC57700.2023.00173.
9. Xing Chen, Jingtao Li, and Chaitali Chakrabarti. "Communication and computation reduction for split learning using asynchronous training". 2021 IEEE Workshop on Signal Processing Systems (SiPS). IEEE. 2021, pp. 76–81.
10. Pengchao Han et. al. "Adaptive Gradient Sparsification for Efficient Federated Learning: An Online Learning Approach". CoRR abs/2001.04756 (2020). arXiv: 2001.04756. URL: <https://arxiv.org/abs/2001.04756>. [23] Sebastian U. Stich et. al. "Sparsified SGD with Memory". CoRR abs/1809.07599 (2018). arXiv: 1809.07599. URL: <http://arxiv.org/abs/1809.07599>.
11. Fei Zheng et. al. Reducing Communication for Split Learning by Randomized Top-k Sparsification. 2023. arXiv: 2305.18469 [cs.LG]. [25]
12. DiChai et. al. "Federated Singular Vector Decomposition". CoRR abs/2105.08925 (2021). arXiv: 2105.08925. URL: <https://arxiv.org/abs/2105.08925>.
13. Suhail Mohamad Shah and Vincent K. N. Lau. "Model Compression for Communication Efficient Federated Learning". IEEE Transactions on Neural Networks and Learning Systems 34.9 (2023), pp. 5937–5951. DOI: 10.1109/TNNLS.2021.3131614.
14. Felix Sattler et. al. "Sparse Binary Compression: Towards Distributed Deep Learning with minimal Communication". CoRR abs/1805.08768 (2018). arXiv: 1805.08768. URL: <http://arxiv.org/abs/1805.08768>.

References



15. Thijs Vogels et. al. "PowerSGD: Practical Low-Rank Gradient Compression for Distributed Optimization". CoRR abs/1905.13727 (2019). arXiv: 1905.13727. URL: <http://arxiv.org/abs/1905.13727>
16. Mahdi Beitollahi and Ning Lu. "FLAC: Federated Learning with Autoencoder Compression and Convergence Guarantee". GLOBECOM 2022- 2022 IEEE Global Communications Conference. 2022, pp. 4589–4594. DOI: 10.1109/GLOBECOM48099.2022.10000743
17. Novoa-Paradela et. al. "Fast deep autoencoder for federated learning". Pattern Recognition 143 (Nov. 2023), p. 109805. ISSN: 0031-3203. DOI: 10.1016/j.patcog.2023.109805. URL: <http://dx.doi.org/10.1016/j.patcog.2023.109805>.
18. Sören Becker et. al. Federated Learning for Autoencoder-based Condition Monitoring in the Industrial Internet of Things. 2022. arXiv: 2211.07619 [cs.LG]
19. Alex Krizhevsky. Learning multiple layers of features from tiny images. Tech. rep. 2009
20. Xiao et al. Fashion-MNIST: a Novel Image Dataset for Benchmarking Machine Learning Algorithms. 2017.
21. Kaggle. APTOS 2019 Diabetic Retinopathy dataset. 2019. URL: <https://www.kaggle.com/competitions/aptos2019-blindness-detection/data>.
22. Kaggle. Kaggle DR dataset (EyePACS). URL: <https://www.kaggle.com/datasets/mariaherrerot/eyepacspreprocess>
23. Ogier du Terrail et. al. "Flamby: Datasets and benchmarks for cross-silo federated learning in realistic healthcare settings". Advances in Neural Information Processing Systems 35 (2022), pp. 5315–5334.
24. Nicholas et. al. Heller. "The state of the art in kidney and kidney tumor segmentation in contrast-enhanced CT imaging: Results of the KiTS19 Challenge". Medical Image Analysis (2020), p. 101821.
25. xitiny dataset. https://torchio.readthedocs.io/datasets.html#torchio_datasets_ixi_ixitiny. Accessed: 2022-05-18.
26. Resnet-18. <https://www.kaggle.com/datasets/pytorch/resnet18>.
27. Margherita Grandini, Enrico Bagli, and Giorgio Visani. "Metrics for Multi-Class Classification: an Overview". ArXiv abs/2008.05756 (2020).
28. Lee Raymond Dice. "Measures of the Amount of Ecological Association Between Species". Ecology 26.3 (July 1945), pp. 297–302. URL: <http://www.istor.org/pss/1932409>
29. <https://paperswithcode.com/dataset/cifar-100>
30. Fabian Isensee et. al. "nnU-Net: self-configuring method for deep learning-based biomedical image segmentation". Nature Methods 18.2 (Feb. 2021), pp. 203–211. ISSN: 1548 7105. DOI: 10.1038/s41467-022-30695-9.
31. Jia Deng et. al. "ImageNet: A large-scale hierarchical image database". IEEE Conference on Computer Vision and Pattern Recognition. 2009, pp. 248–255. DOI: 10.1109/CVPR.2009.5206848. [45]
32. M. Jorge et. al. Cardoso. MONAI Model Zoo. <https://monai.io/model-zoo.html>. Accessed: 2023-06-16.
33. Michela Antonelli et. al. "The Medical Segmentation Decathlon". Nature Communications 13.1 (July 2022), p. 4128. ISSN: 2041-1723. DOI: 10.1038/s41467-022-30695-9. URL: <https://doi.org/10.1038/s41467-022-30695-9>.
34. Michela Antonelli et. al. "The Medical Segmentation Decathlon". Nature Communications 13.1 (July 2022), p. 4128. ISSN: 2041-1723. DOI: 10.1038/s41467-022-30695-9. URL: <https://doi.org/10.1038/s41467-022-30695-9>.
35. Ruoxi Qin et al. "Weighted focal loss: An effective loss function to overcome unbalance problem of chest X-ray14". IOP Conference Series: Materials Science and Engineering. Vol. 428. 1. IOP Publishing. 2018, p. 012022.
36. https://www.sciencedirect.com/science/article/pii/S0925231223009608?dgcid=rss_sd_all

Publication

1. Journal

(a) Under Review- **Manisha Chawla**, Gagan Raj Gupta, Shreyas Gaddam, Manas Wadhwa, "Beyond Federated Learning for IoT: Efficient Split Learning with Caching & Model Customization", IEEE Internet of Things Journal

2. Conference:

(a) Published- Dhruv Deshmukh, Gagan Raj Gupta, **Manisha Chawla**, Vishwesh Jatala, Anirban Haldar, "Entropy Aware Training for Fast and Accurate Distributed GNN", IEEE 2023 IEEE International Conference on Data Mining (ICDM), held on 01– 04 Dec 2023 in Shanghai, China. DOI Bookmark: 10.1109/ICDM58522.2023.00112

(b) Submitted- Aayushman, Hemanth Gaddey, Vidhi Mittal, **Manisha Chawla**, Gagan Raj Gupta, "Fair and Accurate Skin Disease Image Classification using Alignment with Clinical Text Representations", Medical Image Computing and Computer Assisted Intervention (MICCAI) 2024 on March 08, 2024.

Thank You

Use Case:

Uses Cases:

- Finance: Fraud detection, Personalized financial recommendations
- Industry: Predictive maintenance, Product defect detection
- Healthcare: Medical diagnosis

In this study We have mainly focused on healthcare application. Which cover different tasks (classification/segmentation) in multiple application domains and with different data modalities and scale.

All datasets are partitioned using natural splits.

Effect of client back model customization:

	Split	Client Back Side	Number of parameters	CIFAR-10		FMNIST		ISIC		DR	
				ESL-G	ESL-P	ESL-G	ESL-P	ESL-G	ESL-P	ESL-G	ESL-P
small →	RN-S1	LL(512x10)	0.005	82.39	86.65	82.46	<u>91.48</u>	61.90	68.97	50.87	56.50
	C-1	2LL(512x64X10)	0.033	81.65	87.75	82.02	89.72	61.67	69.0	52.9	56.98
Large →	C-5	DS(512x5x5)+LL(512x10)	0.28	<u>84.02</u>	88.36	<u>84.47</u>	90.53	61.13	71.40	53.46	61.30
	RN-S3	Last CB+LL(512x10)	8.39	82.34	<u>88.52</u>	82.55	88.97	<u>66.16</u>	<u>74.43</u>	<u>57.6</u>	<u>63.97</u>

Comparison of performance metric for different configurations of Client Back Models across various splits on various datasets.
The best result is underlined.

Observation:

- By carefully selecting and designing the custom block, it is possible to achieve comparable or even improved performance compared to a larger, non-customized model. This allows for a higher offload of parameters to the server side while maintaining good accuracy and performance.

Conclusion: Customization allows for a higher offload of parameters to the server side while maintaining good accuracy and performance.